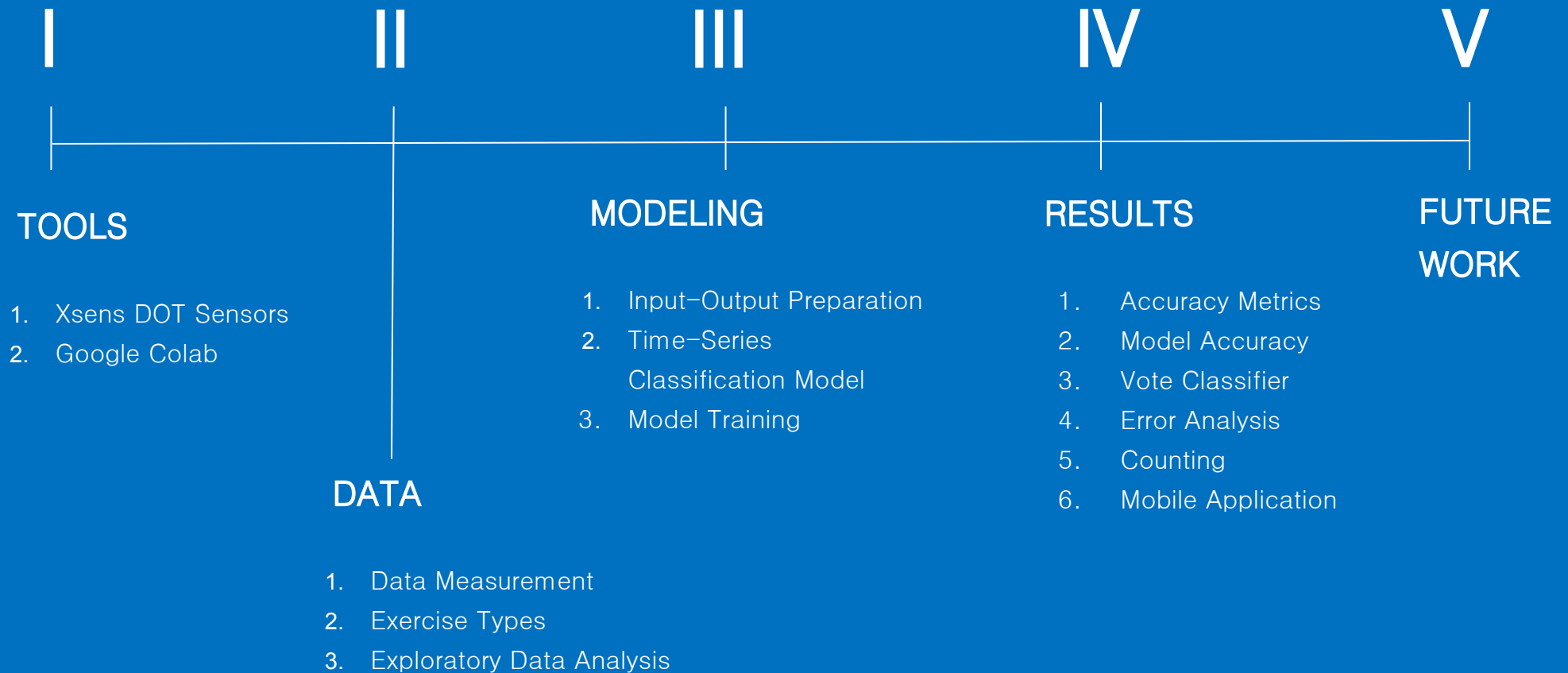


CONTENT



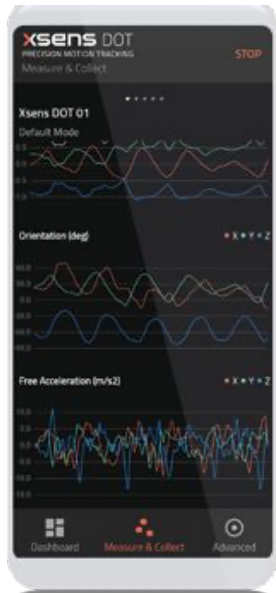
TOOLS

1. Xsens DOT Sensors
2. Setup Google Colab in Your Browser



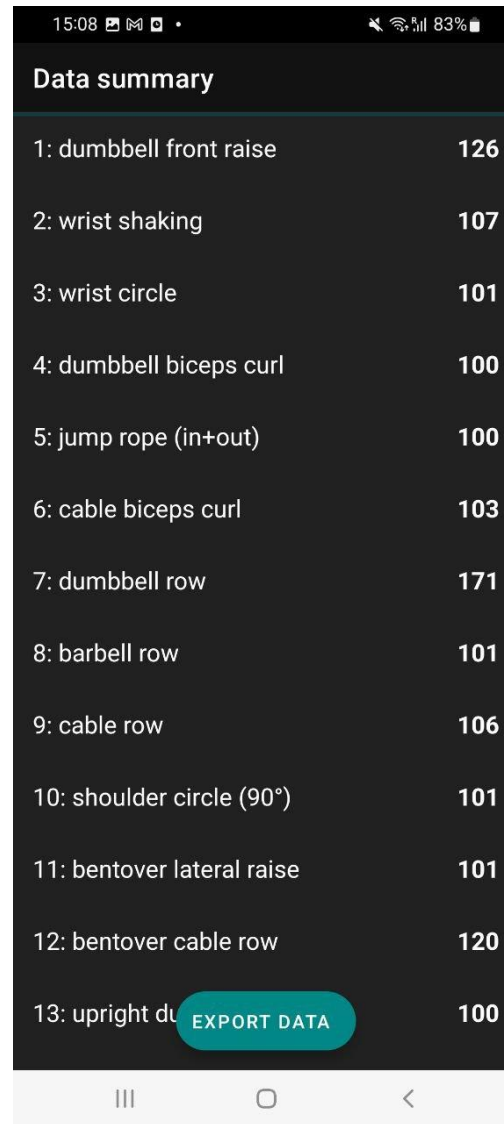
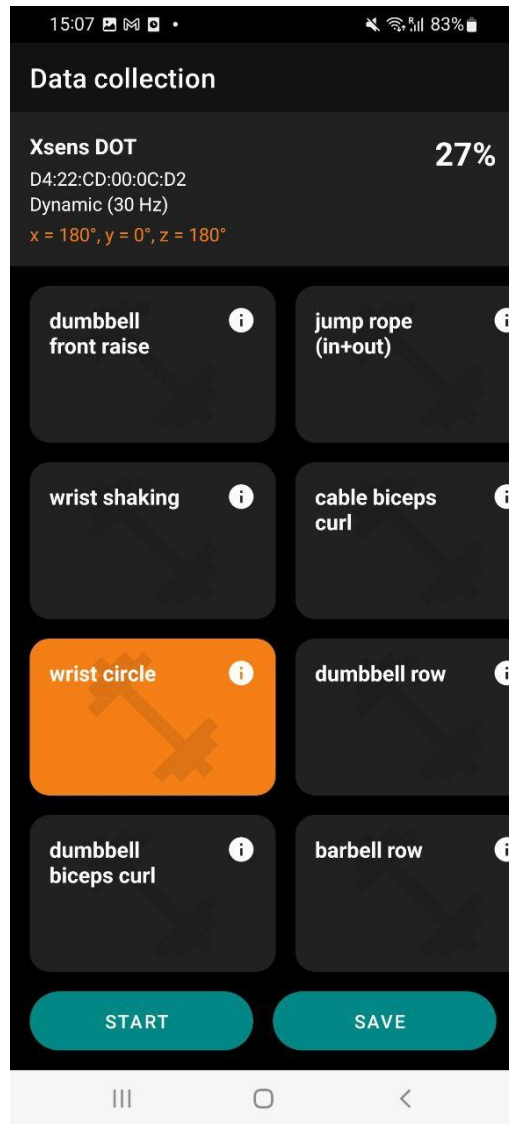
Xsens DOT sensor is small and lightweight for portability, with a quick and easy set up. The IP68 water-resistant rating means a little rain won't stop your progress, and the long battery life gives you enough charge for a day's power.

Connect the DOTs. You can use the Xsens DOT sensor as a standalone device, or you can connect multiple sensors according to your needs for even more accurate complex motion tracking data. The Xsens DOT comes with a slimline charging case for up to 5 sensors, which also keeps them protected when not in use.



Integrate Easily. The Xsens DOT provides endless application possibilities. The platform is flexible, ease-to-integrate with any device, independent of the operating system. So you have everything you need to build a commercially attractive application.

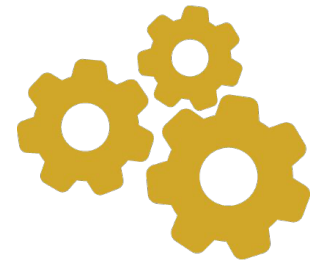
Sensor data collecting app - 40 exercise types and 100 times each





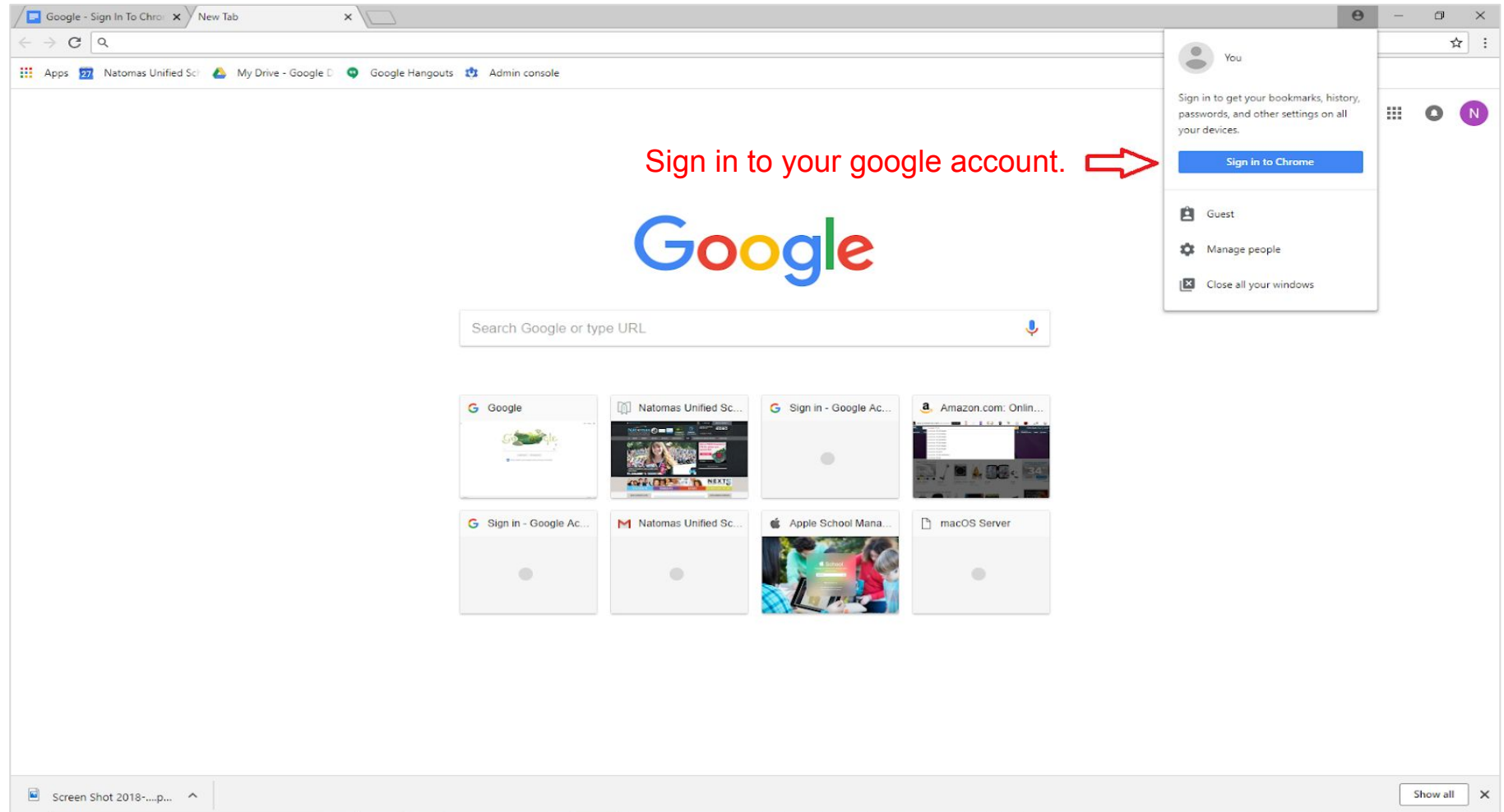
Google is quite aggressive in AI research. Over many years, Google developed AI framework called **TensorFlow** and a development tool called **Colaboratory**. Today TensorFlow is open-sourced and since 2017, Google made Colaboratory free for public use. Colaboratory is now known as **Google Colab** or simply Colab.

Google Colab is a free Jupyter notebook environment that runs entirely in the cloud. Most importantly, it does not require a setup and the notebooks that you create can be simultaneously edited by your team members - just the way you edit documents in Google Docs. Colab supports many popular **machine learning** libraries which can be easily loaded in your notebook.

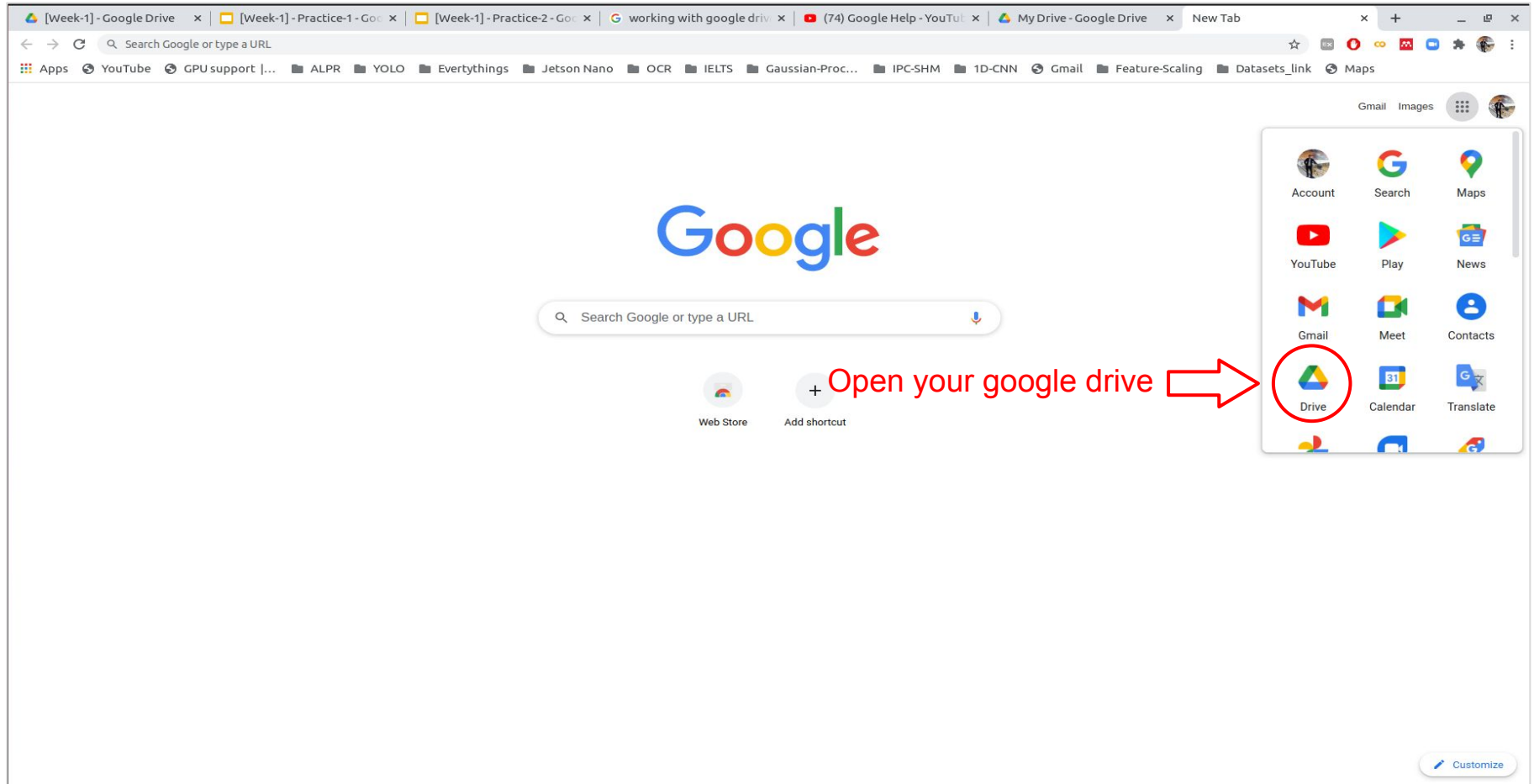


- Write and execute code in **Python**
- Document your code that supports mathematical equations
- Create/Upload/Share notebooks
- Import/Save notebooks from/to Google Drive
- Import/Publish notebooks from GitHub
- Import external datasets e.g. from Kaggle
- Integrate Pytorch, Tensorflow, Keras, etc.
- Free Cloud service with free GPU

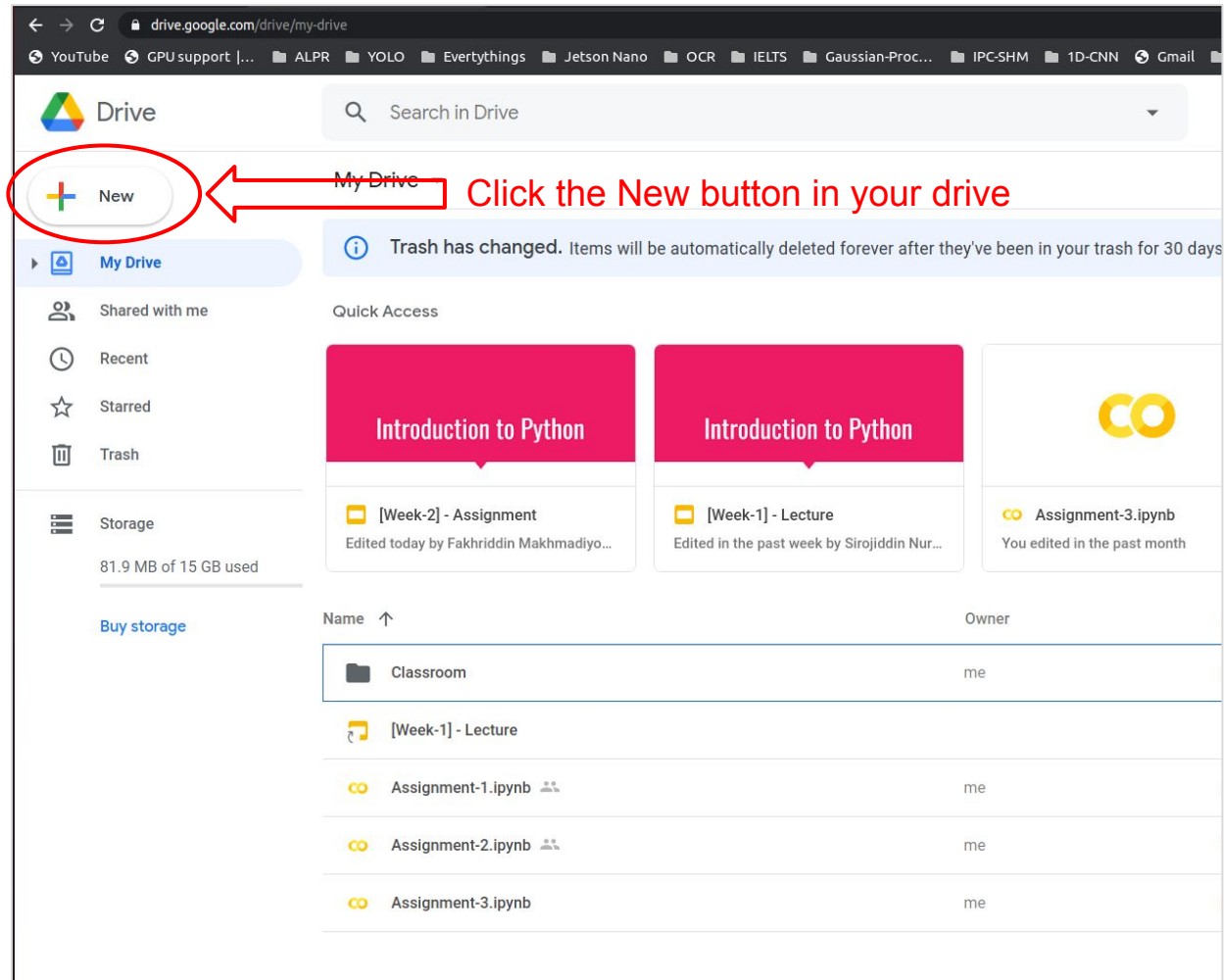
I. TOOLS/ 2. Google Colab



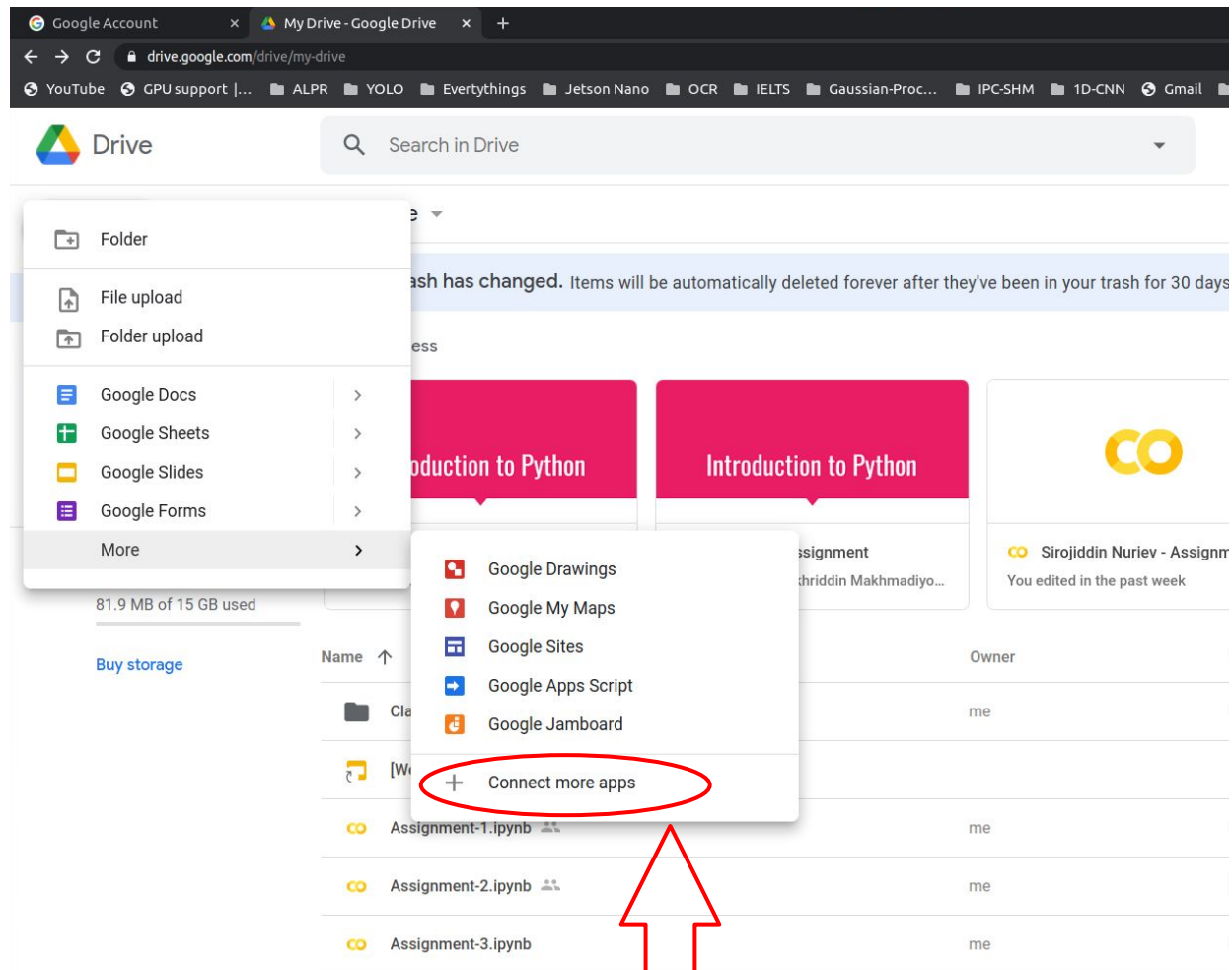
I. TOOLS/ 2. Google Colab



I. TOOLS/ 2. Google Colab

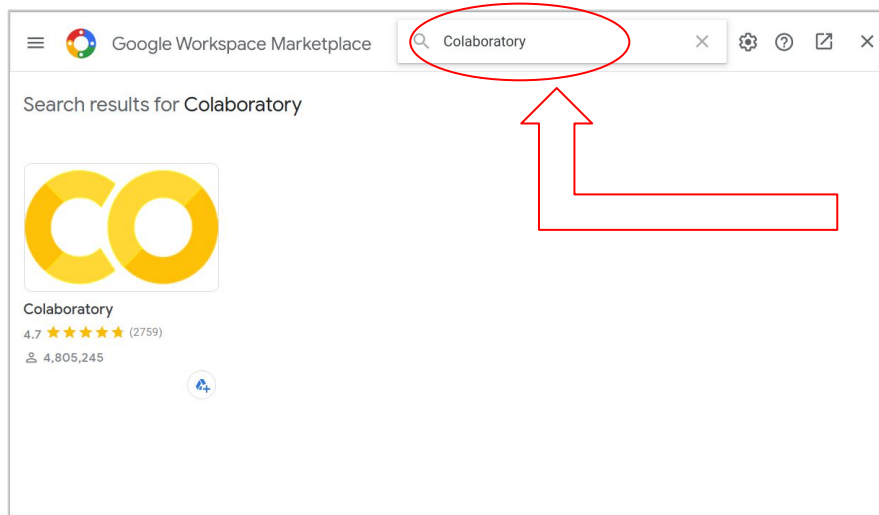


I. TOOLS/ 2. Google Colab

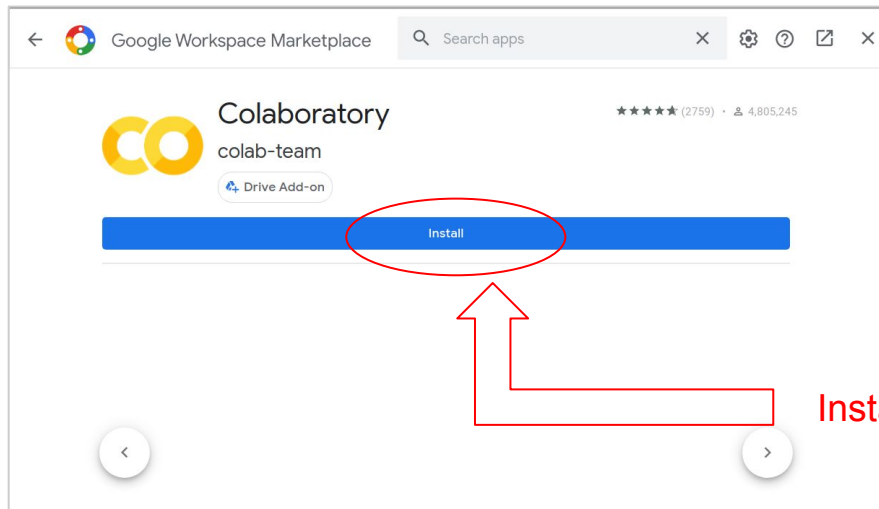


If you don't see “Google Colaboratory” at the top, click Connect more apps

I. TOOLS/ 2. Google Colab



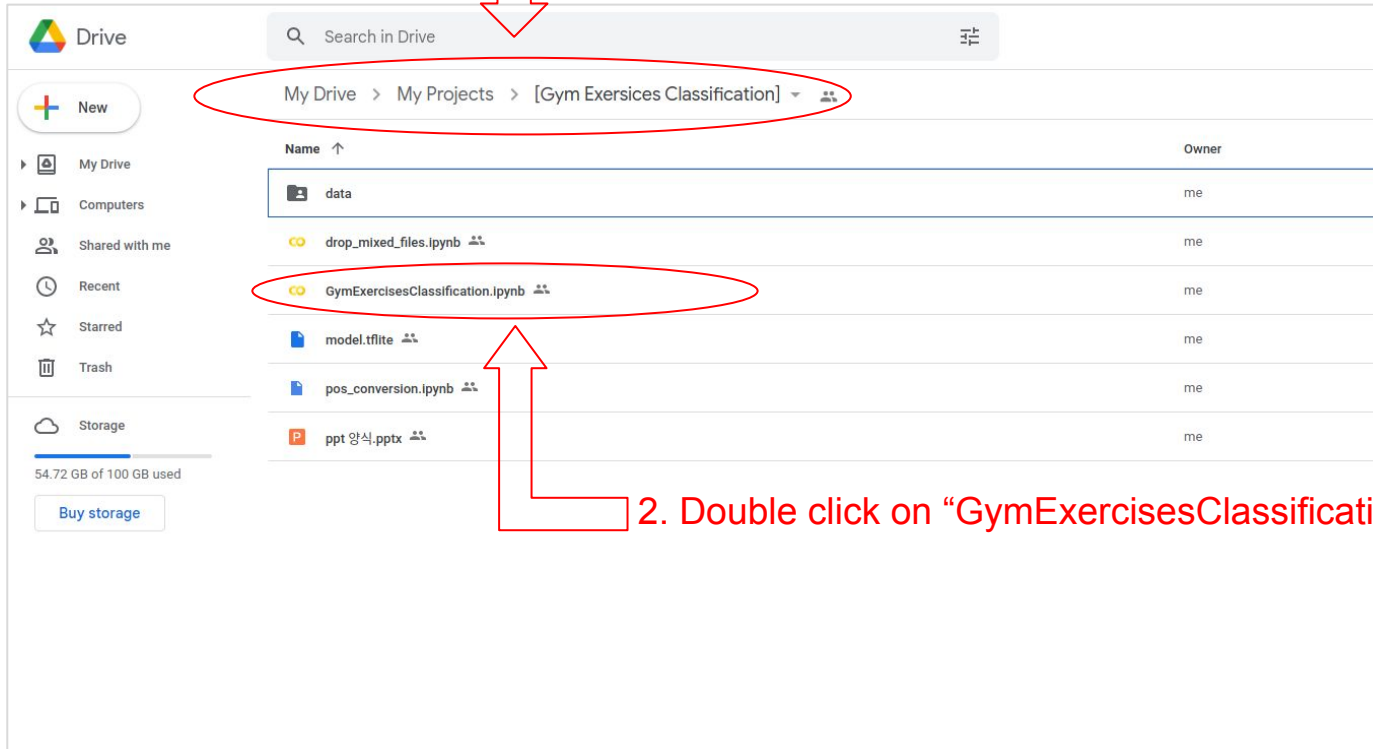
Search “Colaboratory” extension.



Install “Colaboratory” extension.

I. TOOLS/ 2. Google Colab

1. Upload the “Gym Exercises Classification” folder to your google drive



2. Double click on “GymExercisesClassification.ipynb” colab file

I. TOOLS/ 2. Google Colab

The screenshot shows a Google Colab notebook interface. The browser address bar displays the URL: `colab.research.google.com/drive/1teBNe6pY0tKkjjWwNgq0Jvc9Ibeg96MRZ#scrollTo=2CQRutOV0Bq`. The notebook title is `Gym Exercises Classification.ipynb`. The left sidebar contains a 'Table of contents' panel with the following sections:

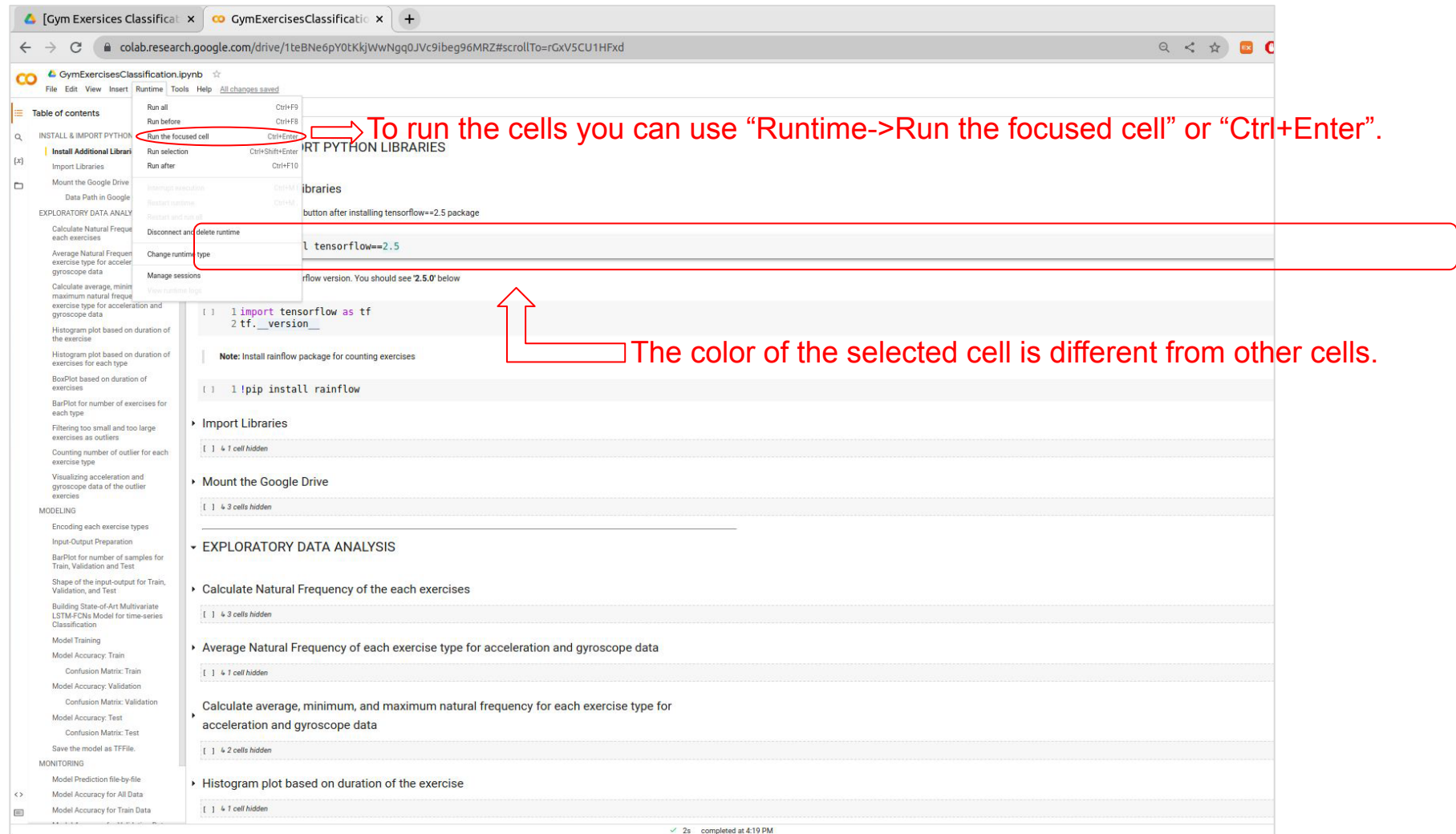
- INSTALL & IMPORT PYTHON LIBRARIES
 - Install Additional Libraries
 - Import Libraries
 - Mount the Google Drive
 - Data Path in Google Drive
- EXPLORATORY DATA ANALYSIS
 - Calculate Natural Frequency of the each exercises
 - Average Natural Frequency of each exercise type for acceleration and gyroscope data
 - Calculate average, minimum, and maximum natural frequency for each exercise type for acceleration and gyroscope data
 - Histogram plot based on duration of the exercise
 - Histogram plot based on duration of exercises for each type
 - BoxPlot based on duration of exercises
 - BarPlot for number of exercises for each type
 - Filtering too small and too large exercises as outliers
 - Counting number of outlier for each exercise type
 - Visualizing acceleration and gyroscope data of the outlier exercises
- MODELING
 - Encoding each exercise types
 - Input-Output Preparation
 - BarPlot for number of samples for Train, Validation and Test
 - Shape of the input-output for Train, Validation, and Test
 - Building State-of-Art Multivariate LSTM-FCNs Model for time-series Classification
 - Model Training
 - Model Accuracy: Train
 - Confusion Matrix: Train
 - Model Accuracy: Validation
 - Confusion Matrix: Validation
 - Model Accuracy: Test
 - Confusion Matrix: Test
 - Save the model as TFFile
- MONITORING
 - Model Prediction file-by-file
 - Model Accuracy for All Data
 - Model Accuracy for Train Data

The main notebook area displays the following tasks:

- INSTALL & IMPORT PYTHON LIBRARIES
 - Install Additional Libraries
 - Note: Click the restart button after installing tensorflow==2.5 package
 - 4 5 cells hidden
 - Import Libraries
 - 4 1 cell hidden
 - Mount the Google Drive
 - 4 3 cells hidden
- EXPLORATORY DATA ANALYSIS
 - Calculate Natural Frequency of the each exercises
 - 4 3 cells hidden
 - Average Natural Frequency of each exercise type for acceleration and gyroscope data
 - 4 1 cell hidden
 - Calculate average, minimum, and maximum natural frequency for each exercise type for acceleration and gyroscope data
 - 4 2 cells hidden
 - Histogram plot based on duration of the exercise
 - 4 1 cell hidden
 - Histogram plot based on duration of exercises for each type
 - 4 1 cell hidden
 - BoxPlot based on duration of exercises
 - 4 1 cell hidden
 - BarPlot for number of exercises for each type

The bottom status bar indicates: `2s completed at 4:19 PM`.

1. TOOLS/ 2. Google Colab



The screenshot shows the Google Colab interface for a file named 'GymExercisesClassification.ipynb'. The 'Runtime' menu is open, showing options like 'Run all', 'Run before', 'Run the focused cell', 'Run selection', 'Run after', 'Interrupt execution', 'Restart runtime', 'Disconnect and delete runtime', 'Change runtime type', and 'Manage sessions'. A red arrow points from the text 'To run the cells you can use "Runtime->Run the focused cell" or "Ctrl+Enter"' to the 'Run the focused cell' option. Another red arrow points from the text 'The color of the selected cell is different from other cells.' to a cell containing the code 'import tensorflow as tf' and 'tf.__version__'. A red box highlights the 'Change runtime type' option in the 'Runtime' menu.

To run the cells you can use "Runtime->Run the focused cell" or "Ctrl+Enter".

The color of the selected cell is different from other cells.

Follow the instructions in the Colab file!

I. TOOLS/ 2. Google Colab

2 After mounting your drive you can see the all files in your drive as shown on the left.

1 Mount the Google Drive

```
1 from google.colab import drive
2 drive.mount('/content/drive')
```

Mounted at /content/drive

3

```
1 data_folder = '/content/drive/MyDrive/My Projects/[Gym Exercises Classification]/data/'
2 features = ['accX', 'accY', 'accZ', 'orientationX', 'orientationY', 'orientationZ']
```

2

Copy path

```
pd.DataFrame(columns = features+['type', 'duration'])
gym in sorted(os.listdir(data_folder)):
    file in sorted(os.listdir(data_folder+type_gym)):
        f = pd.read_csv(data_folder+type_gym+'/'+file)
        s = 30
        = df[features].values
        = np.linspace(0, len(x)/f.s, len(x), endpoint=False)
        freq = np.abs(fftpack.fft(x, axis=0))
        reqs = fftpack.fftfreq(len(x))*f.s
        try:
            df.freq.loc[file] = list(np.abs(freqs[(abs(freqs)>0.1) & (abs(freqs)<3)][X_freq[(abs(freqs)>0.1) & (abs(freqs)<3)].argmax(axis=0)])) + [type_gym, len(x)]
        except:
            print(type_gym, file)
            print(df.info())
        df_freq.head()
```

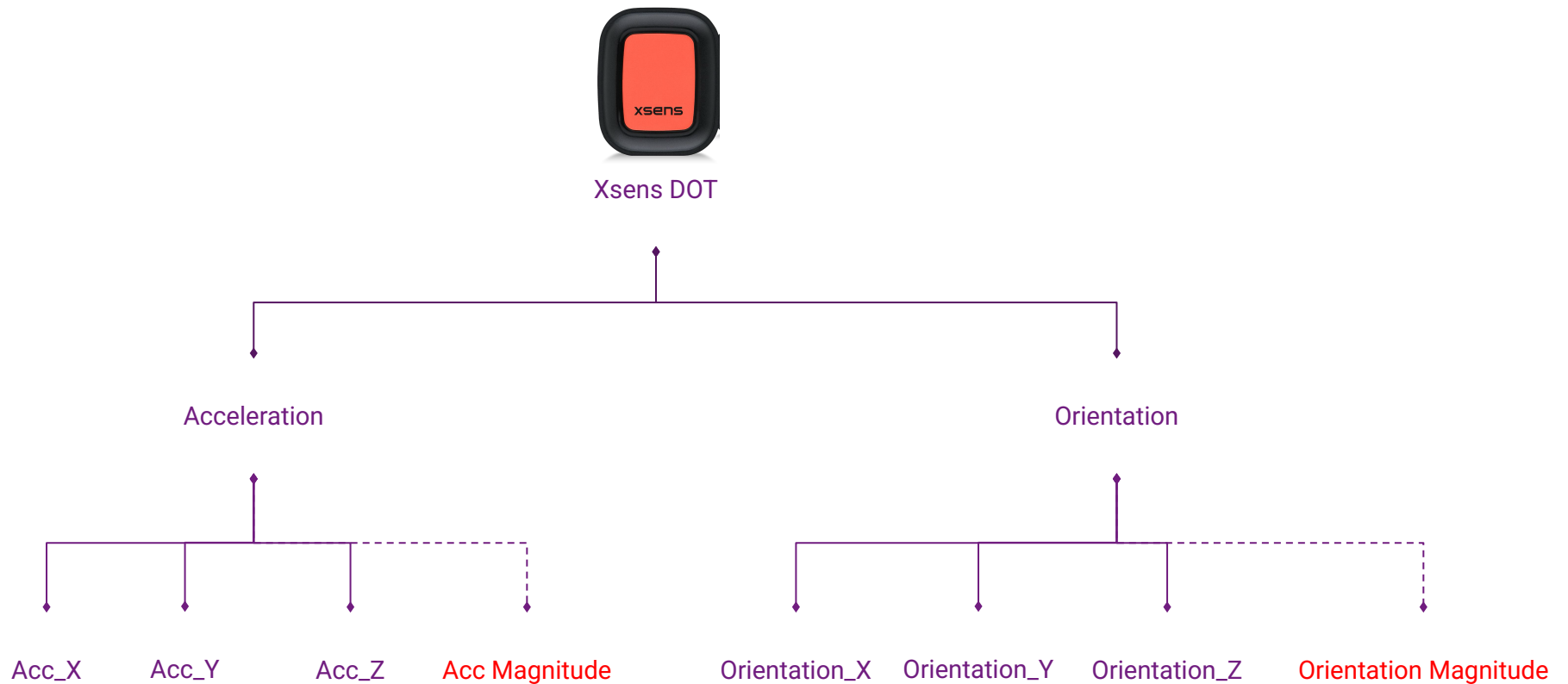
	accX	accY	accZ	orientationX	orientationY	orientationZ	type	duration
0FVVRFOqpnMEXMYBgKRO.csv	1.962617	1.962617	1.121495	1.121495	0.280374	1.962617	arm scissor(front)	107
0eLTxyNhwCZR5q1LZE.csv	2.264151	1.698113	2.547170	0.849057	0.566038	1.698113	arm scissor(front)	106
1PZBKDU21DoVyPT2ZFjd.csv	2.117647	2.117647	1.058824	1.058824	2.117647	2.117647	arm scissor(front)	85
1vH2rz9y9KIMNagDw.csv	2.168675	2.168675	2.168675	1.084337	2.168675	2.168675	arm scissor(front)	83
36EPVEZTwqTCJXJBVBrr.csv	2.187500	2.187500	2.187500	0.937500	0.312500	1.875000	arm scissor(front)	96

21s completed at 8:54 PM

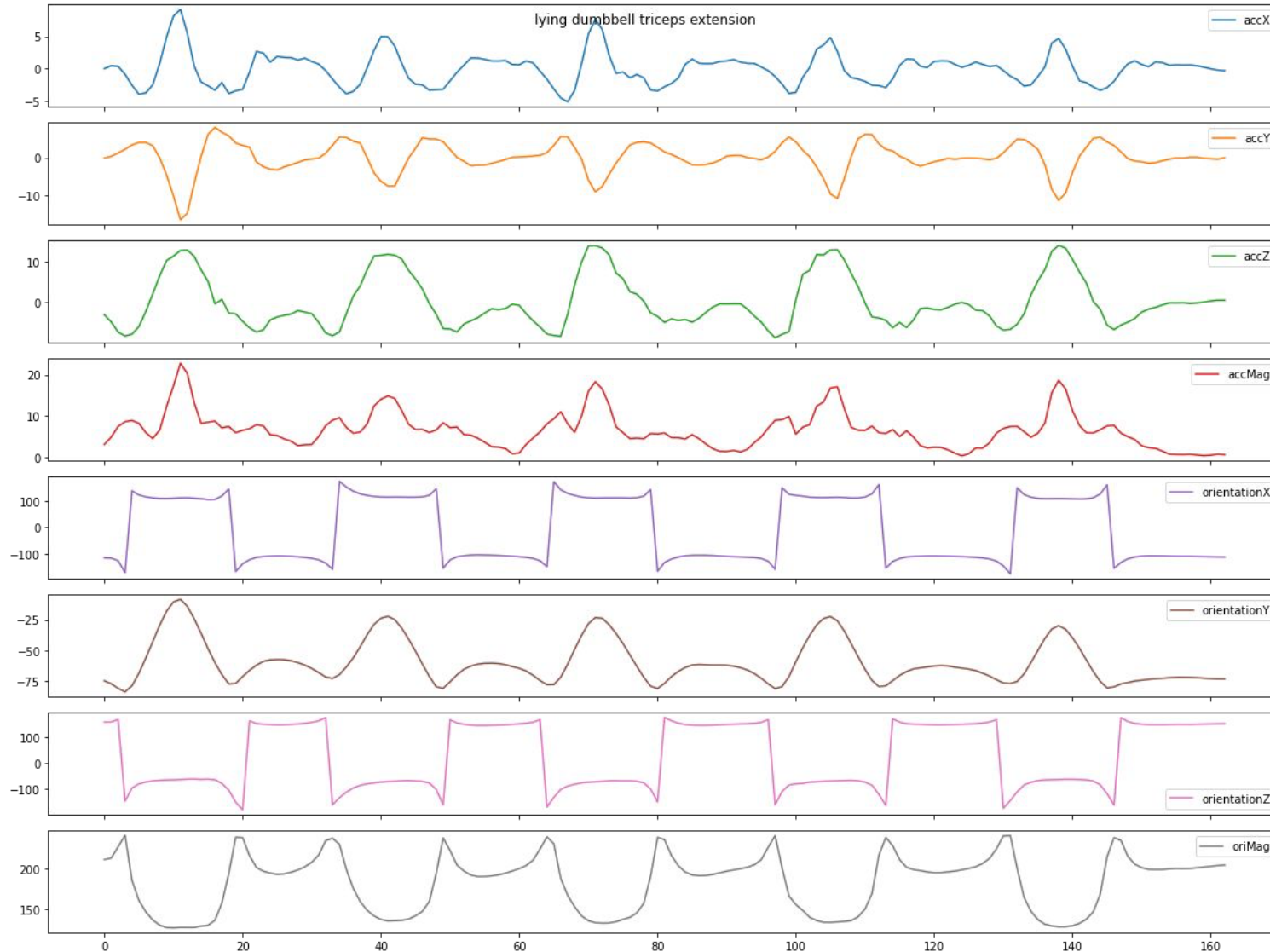
DATA

- 1 Data Measurement
- 2 Types of Exercises
- 3 Exploratory Data Analysis

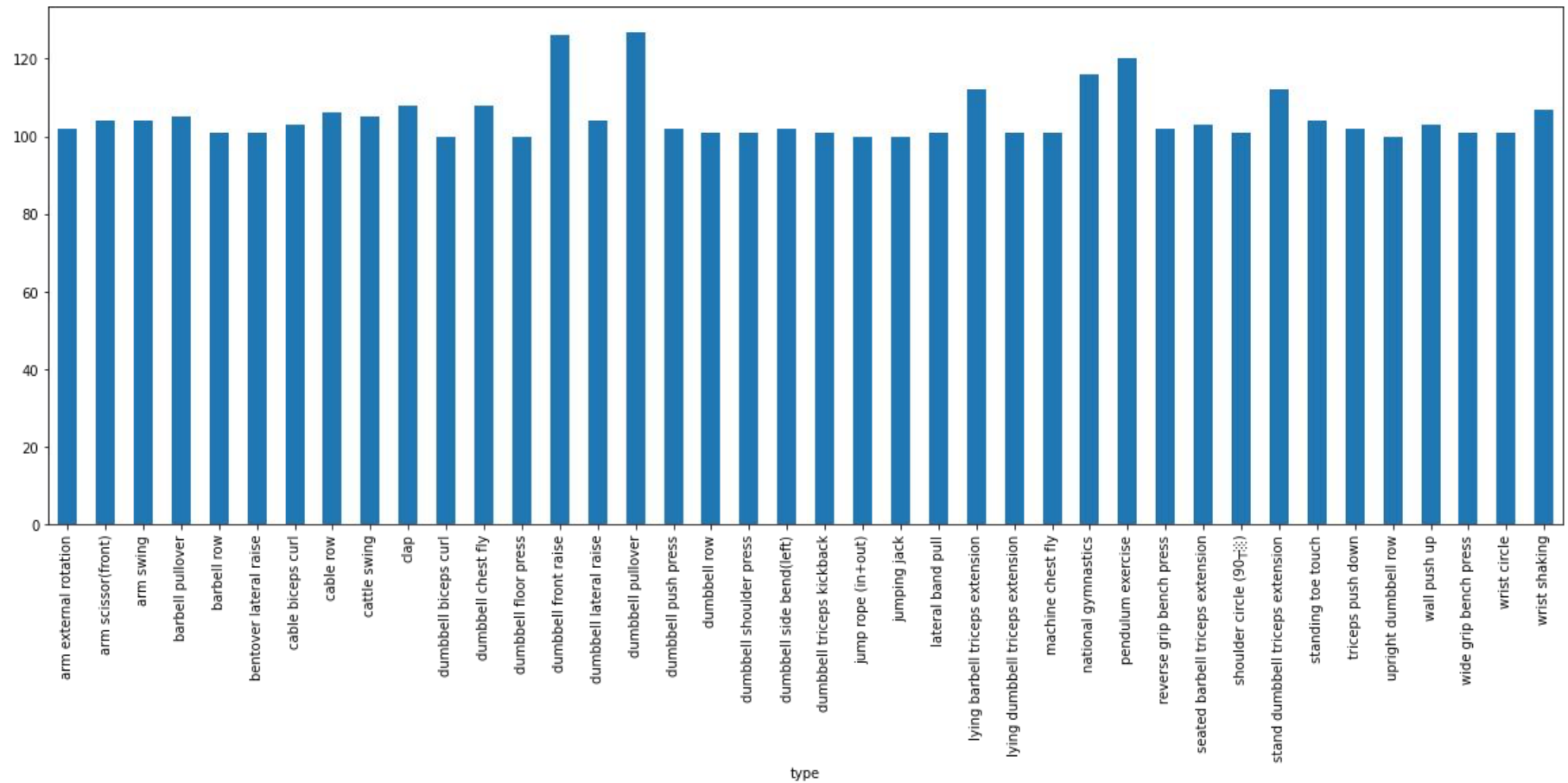




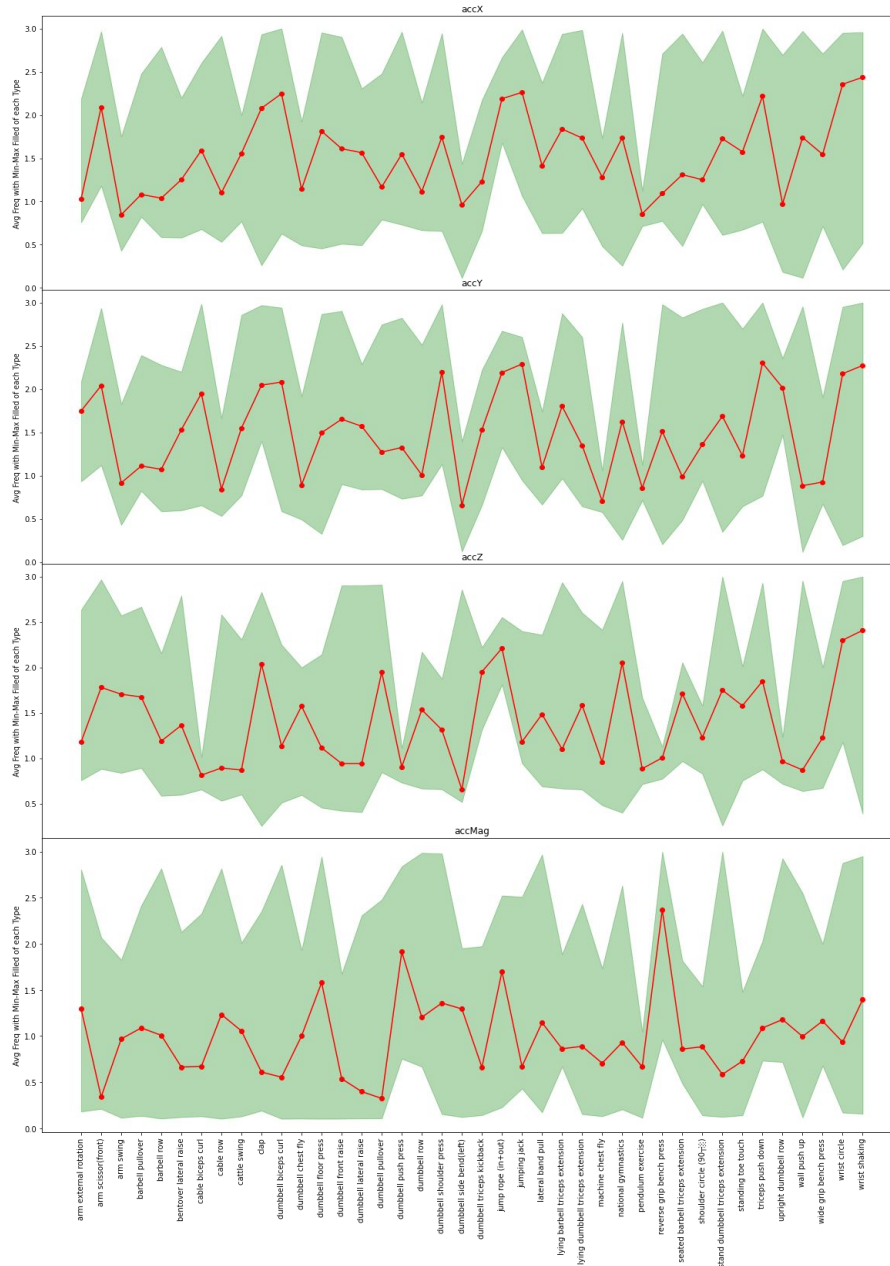
The sensor is configured to measure at 30 Hz. This means that the sensor measures 3D acceleration and 3D orientation 30 times per second. After magnitude of acceleration is calculated, you receive a total of 30 x 8 data per second.



This plot shows 5 sets of the **lying dumbbell triceps extension** exercise in about 4 seconds. The peak frequency represents each set of the exercise in frequency domain.



There are 40 types of exercises with about 100 samples each. Each exercise sample was saved in separate csv files in a folder named with the exercise type. And each exercise was done 5 sets.



The peak frequency was calculated for each sample by converting the data from the time domain to the frequency domain.

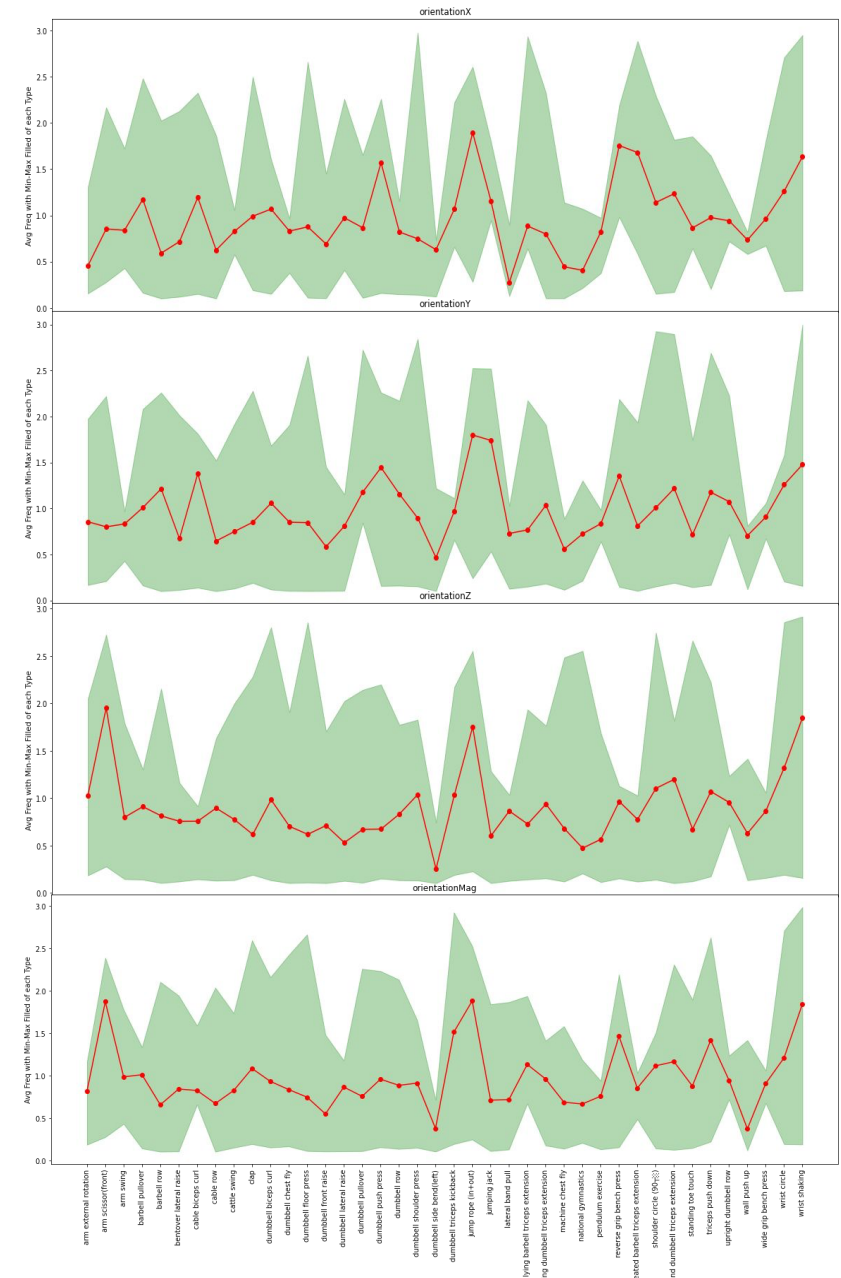
In this plot, the red line represents the average value of the peak frequencies of samples of the same exercise type for each axis of the acceleration data. And the filled area with green shows the smallest and largest peak frequencies of that type of exercise.

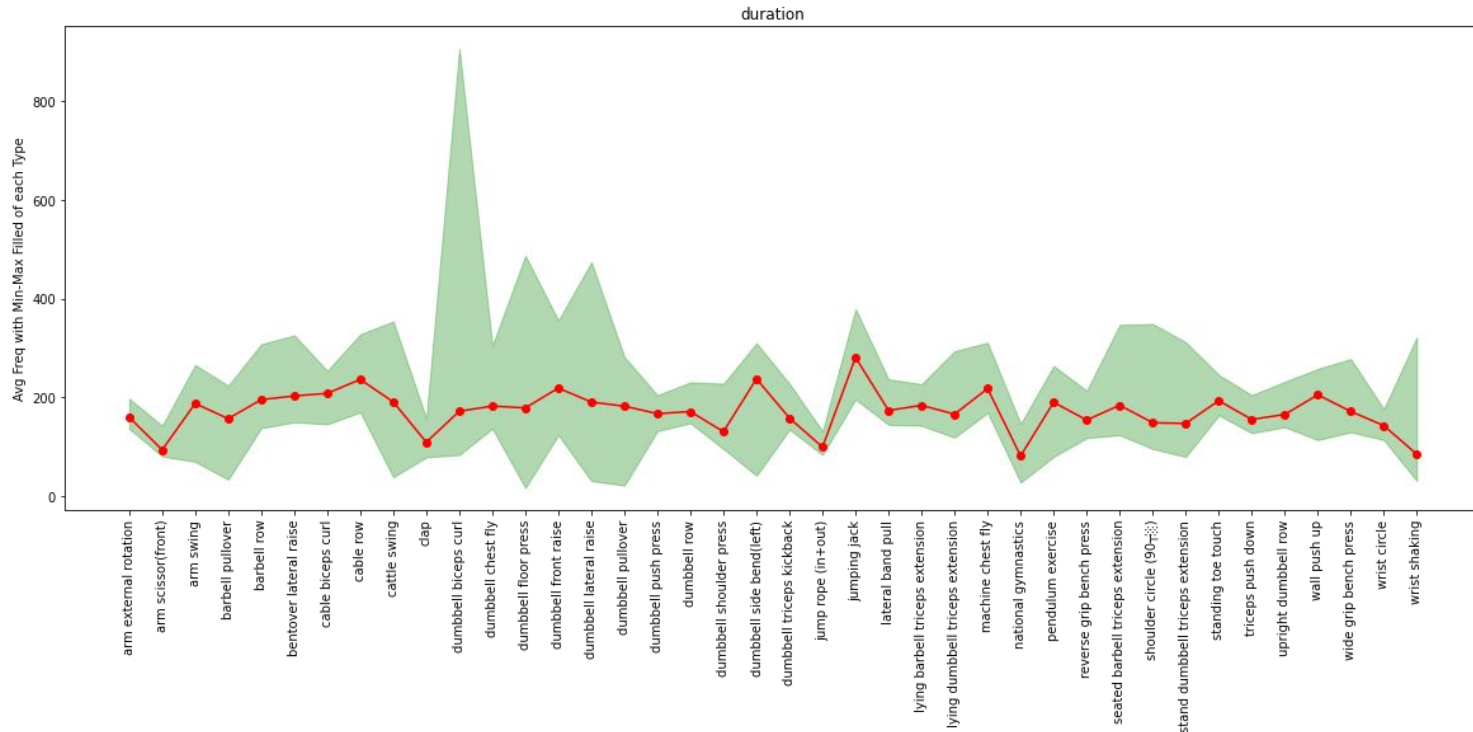
The difference between the maximum and minimum peak frequencies indicates how different the set durations of same type of exercise samples.

The peak frequency was calculated for each sample by converting the data from the time domain to the frequency domain.

In this plot, the red line represents the average value of the peak frequencies of samples of the same exercise type for each axis of the orientation data. And the area filed with green shows the smallest and largest peak frequencies of that type of exercise.

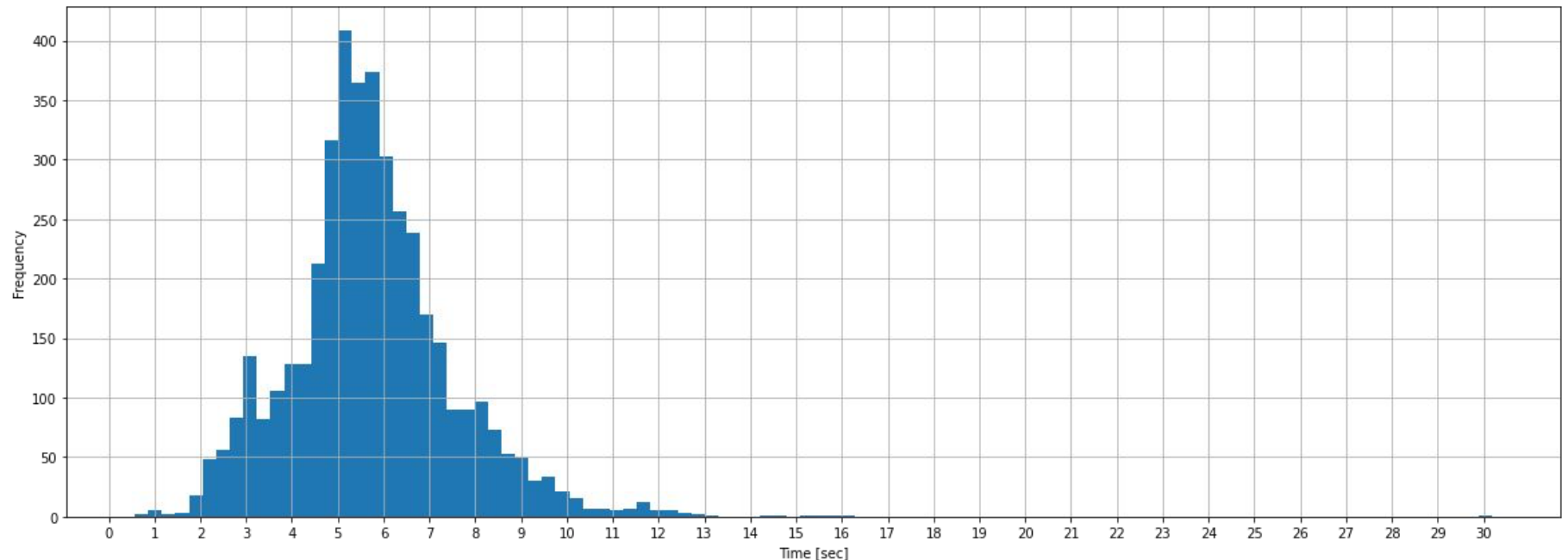
The difference between the maximum and minimum peak frequencies indicates how different the set durations of same type of exercise samples.





As we mentioned above, each exercise was done in 5 sets. In this plot, the duration of each exercise was compared by type. The red line represents the average time spent on completing 5 sets of each type of exercise. The area filled in green shows the minimum and maximum time spent on completing 5 sets of each exercise type.

In this visualization, it can be seen that the time taken to complete 5 sets of exercises for some types varies greatly from one to another. For example, it took at least 4 seconds to complete 5 sets, and at most 30 seconds in the "**dumbbell biceps curl**" type. This difference indicates there are some outlier samples in this type of exercise.

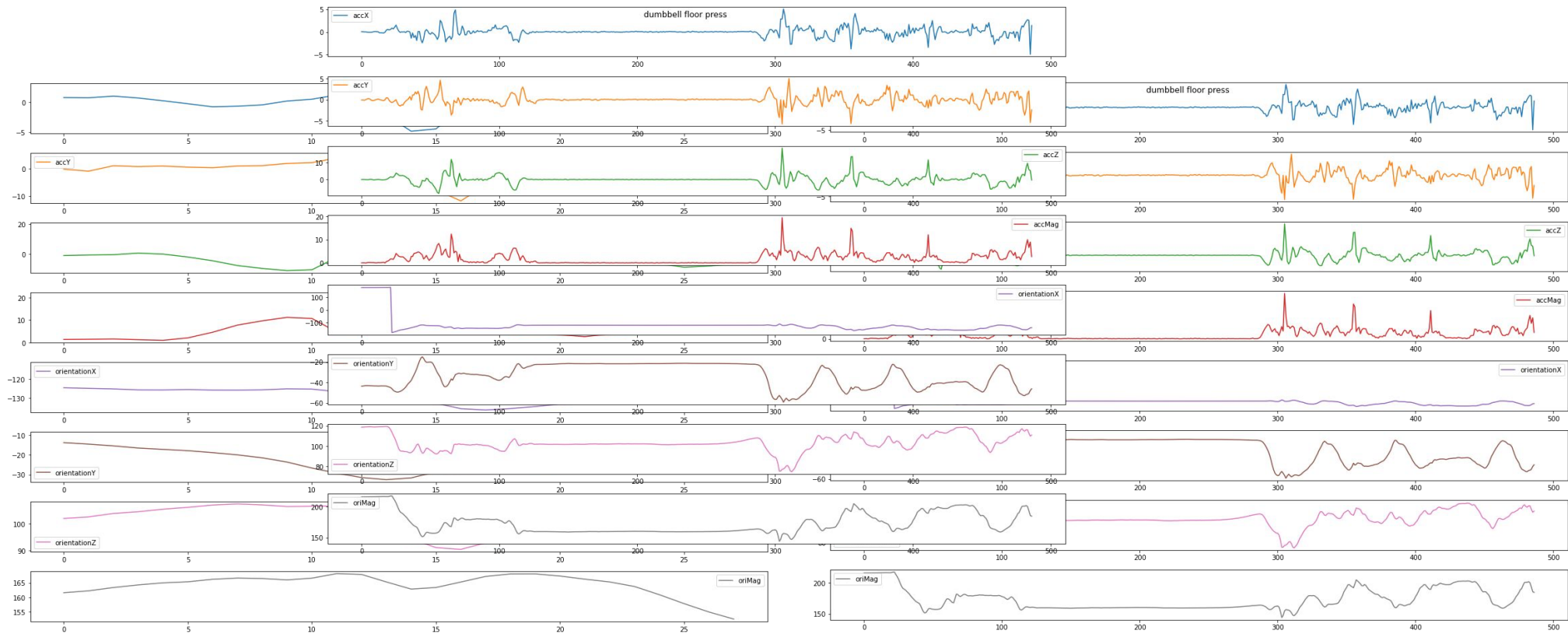


The histogram plot shows the frequency distribution of duration of each samples. There are few exercises complete the 5 sets less than 1.5 seconds and same more than 14 seconds. Because doing 5 sets in less than 1.5 seconds is not easy. Users may have done more than 5 sets or started later after pressing the start button on the sensor or paused between sets if they spent more than 14 seconds. We filtered them out as an **outlier**.

index	type	duration
54rDjlk0C6gJ0plg4ipR.csv	dumbbell floor press	17
cuTsMZwuJjC5qwKOZjta.csv	dumbbell pullover	22
1s4fO6gRwdWkGGHRme2A.csv	dumbbell floor press	28
j0bppfOMGaT4pTrkhKPD.csv	national gymnastics	28
WGxTccOglOZLMEJICdRq.csv	dumbbell lateral raise	31
oGBYSnqSisdqxG2dru3F.csv	wrist shaking	31
jJZi2uj9BTDn44nF29Zc.csv	barbell pullover	34
mgq8ewAbaD1v9rFwbGQY.csv	cattle swing	39
j3z4C5pjXqOuCsW5Khlc.csv	dumbbell side bend(left)	42
GQz5Zzig3CenuWAUIxV6.csv	dumbbell lateral raise	429
mhzyNdD63A9flxDNleSj.csv	dumbbell lateral raise	442
oZ6pxvbVSCp2O3VjJoHY.csv	dumbbell floor press	461
kjecAki9HqRY5rAOtPCj.csv	dumbbell lateral raise	468
9dlrXfZSEuNE0W184Q4A.csv	dumbbell lateral raise	474
1Fe8BF5nhZ38cFTRwuSi.csv	dumbbell floor press	487
erfZwZILRSXgXzMsuLgT.csv	dumbbell biceps curl	906

	count
type	
barbell pullover	1
cattle swing	1
dumbbell biceps curl	1
dumbbell floor press	4
dumbbell lateral raise	5
dumbbell pullover	1
dumbbell side bend(left)	1
national gymnastics	1
wrist shaking	1

Here, 5 samples from the "**dumbbell floor press**" and 4 samples from the "**dumbbell lateral raise**" were found as an outlier. It is shown in the first table that some samples' duration is less than 1.5 seconds and some other samples' duration of the same exercise type is more than 14 seconds to complete 5 sets.



On the left, it took less than a second to complete 5 sets of the “**dumbbell floor press**” but left plot shows it is not 5 sets. On the right, completing 5 sets of the same type of exercise took almost 17 seconds and right plot shows there is a pause in between sets. These samples reduce the learning ability of the exercise classification deep learning model. Therefore, we decided to filter out them as an outliers.

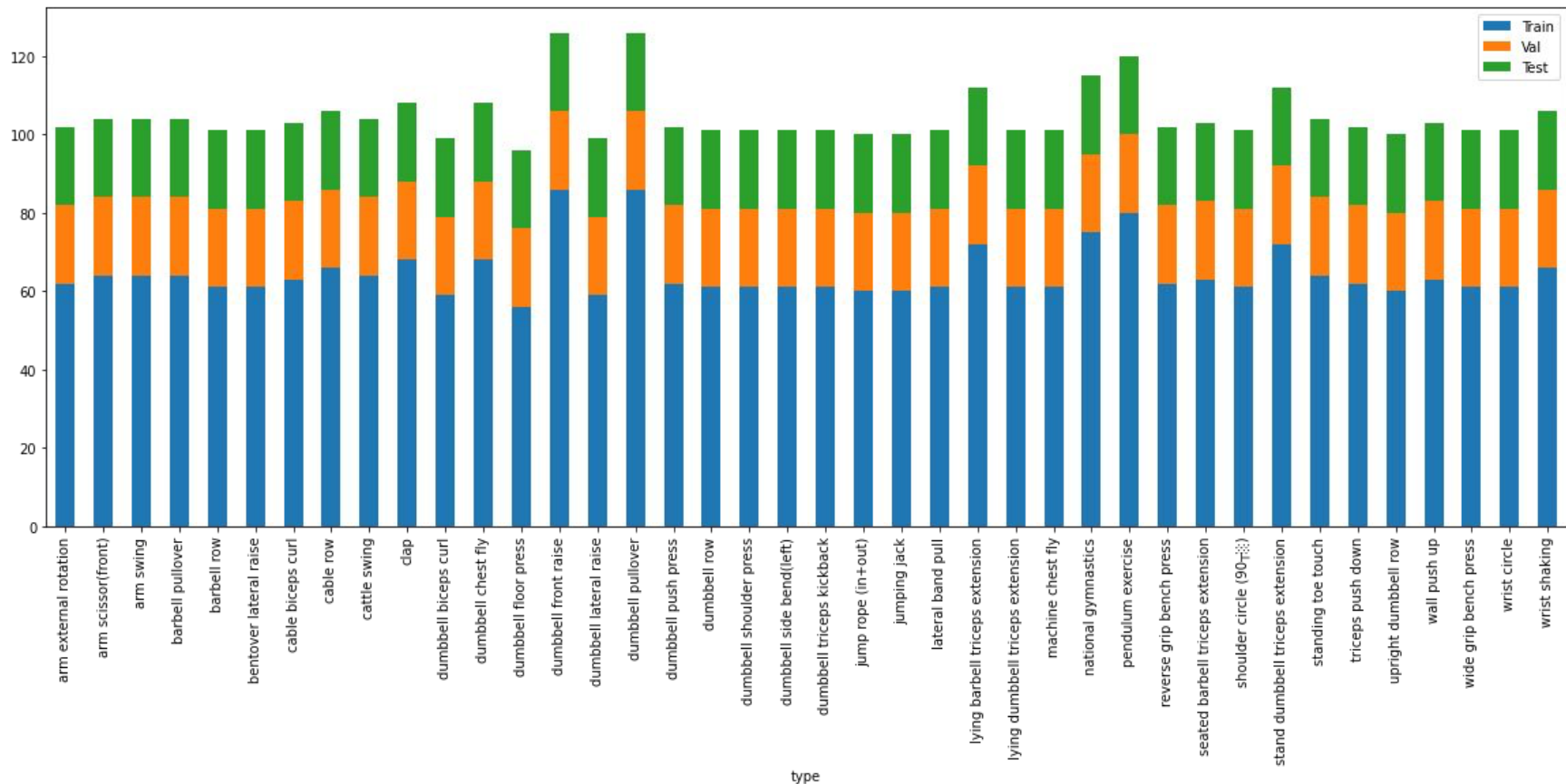
MODELING

1. Input-Output Preparation
2. Time-Series Classification Model
3. Model Training

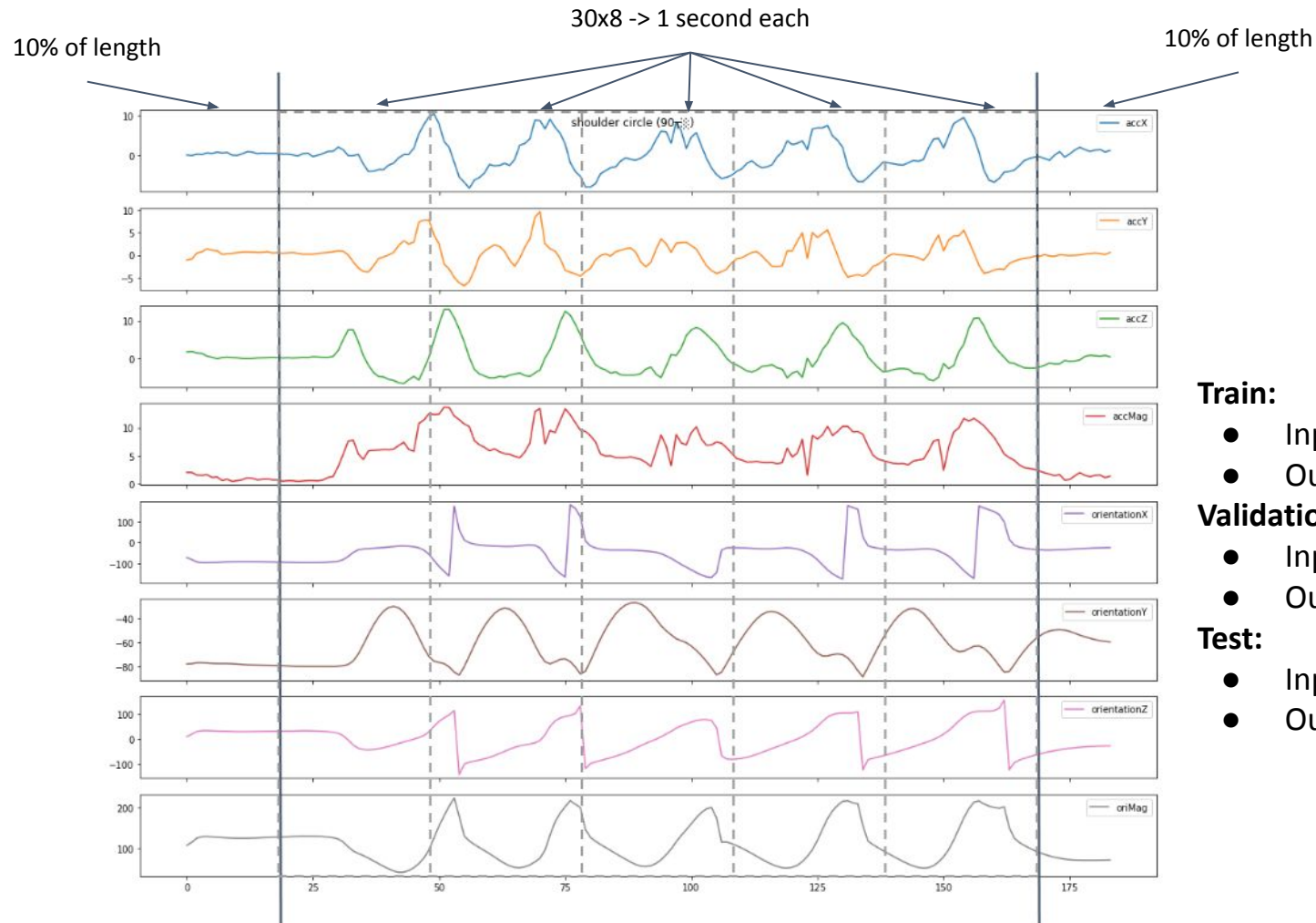




We have about 100 samples for each exercise types. So, we split random 20 samples as validation data and random 20 samples as test data, and remaining as train data for each type.



There were some pause at the both end of the most samples. Therefore, we cut 10% from the beginning and at the end of the each samples. Then, we split every second 30x8 data as an input for the classification model and type of the exercise as an output.



Train:

- Input shape: (10683, 30, 8)
- Output shape: (10683,)

Validation:

- Input shape: (3297, 30, 8)
- Output shape: (3297,)

Test:

- Input shape: (3305, 30, 8)
- Output shape: (3305,)

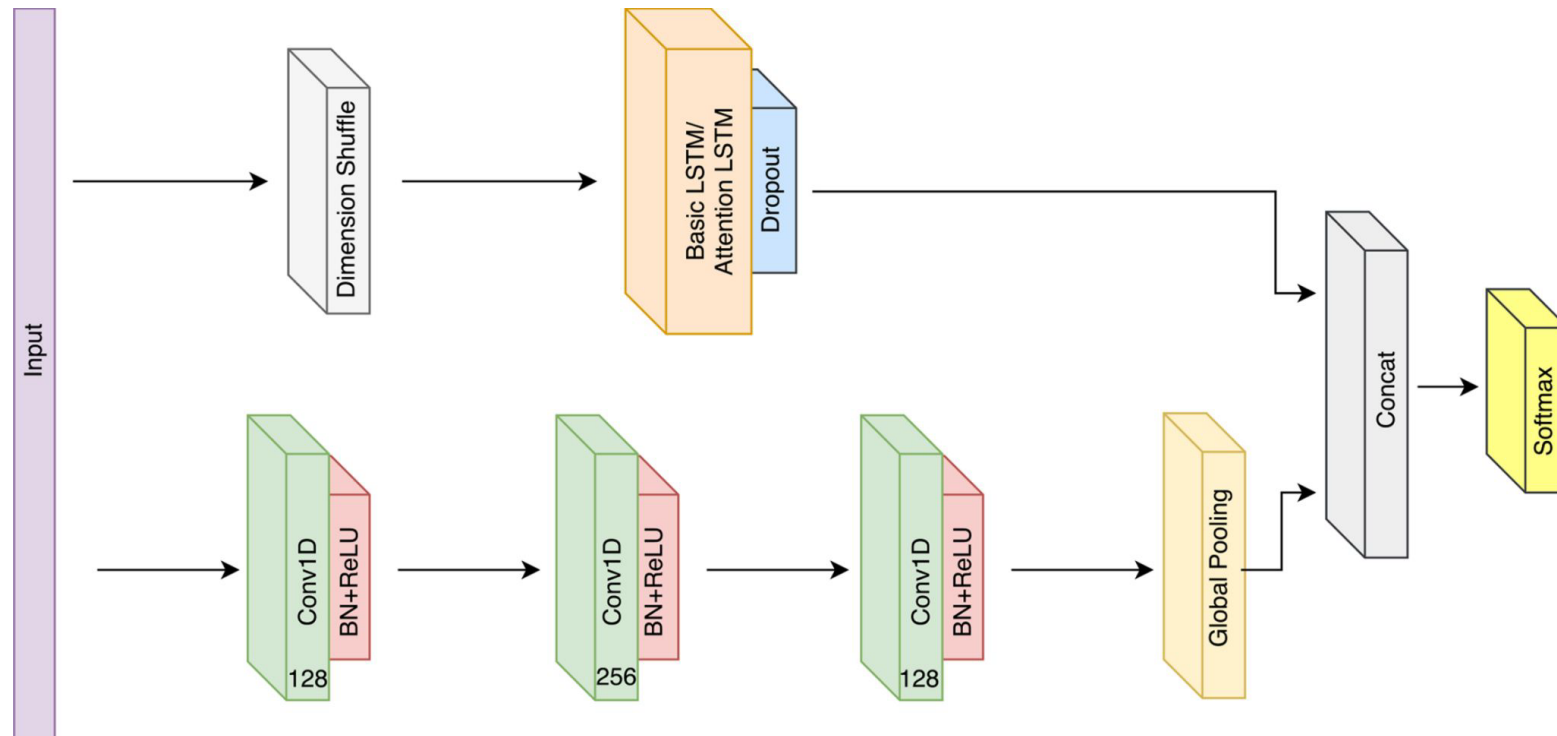
We have used one of the state-of-art time-series classification models called "MLSTM-FCN".

Benchmarks Add a Result

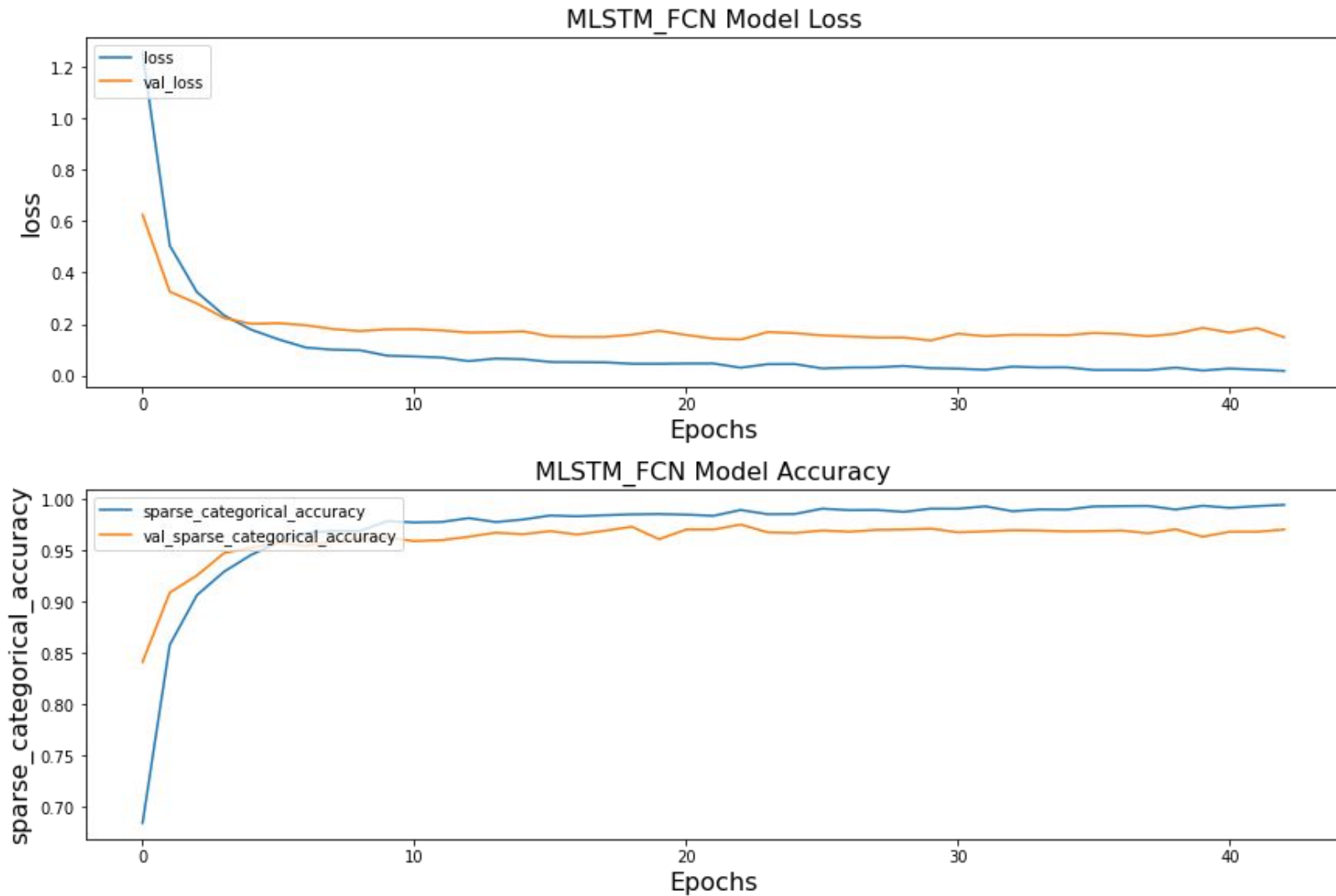
These leaderboards are used to track progress in Time Series Classification

Trend	Dataset	Best Model	Paper	Code	Compare
	PhysioNet Challenge 2012	🏆 GRU-D			See all
	ArabicDigits	🏆 GP-Sig-GRU			See all
	AUSLAN	🏆 FCN-SNLST			See all
	CharacterTrajectories	🏆 MALSTM-FCN			See all
	CMUsubject16	🏆 GP-Sig-GRU			See all
	DigitShapes	🏆 GP-Sig-LSTM			See all
	ECG	🏆 MALSTM-FCN			See all
	JapaneseVowels	🏆 MALSTM-FCN			See all
	KickvsPunch	🏆 MALSTM-FCN			See all
	Libras	🏆 MALSTM-FCN			See all

This screenshot is taken from <https://paperswithcode.com/task/time-series-classification>.



Multivariate LSTM-FCN for Time Series Classification Model Architecture



The model is trained 100 epochs with 32 batch size and achieved over 99% training accuracy and over 98% validation accuracy.

RESULTS

- 1 Accuracy Metrics
- 2 Model Accuracy
- 3 Voting
- 4 Error Analysis
- 5 Counting
- 6 Mobile Application

IV

To check trained model accuracy, we have used Confusion Matrix, Precision, Recall, and F1-Score. What are these metrics?

- **Confusion matrix** is a specific table that allows visualization of the performance of the classification model. Each row of the matrix represents the instances in an actual class while each column represents the instances in a predicted class as shown in the table below.

		Predicted Class	
		Positive	Negative
Actual Class	Positive	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

- **Precision** calculates that what proportion of model prediction was actually correct?
$$\text{Precision} = \frac{TP}{TP + FP}$$
- **Recall** calculates that what proportion of actual class was predicted correctly?
$$\text{Recall} = \frac{TP}{TP + FN}$$
- **F1-Score** is the harmonic mean of the precision and recall.
$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}}$$

Train Accuracy

Train Data Confusion Matrix																																										precision	recall	f1-score	support																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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Validation Accuracy

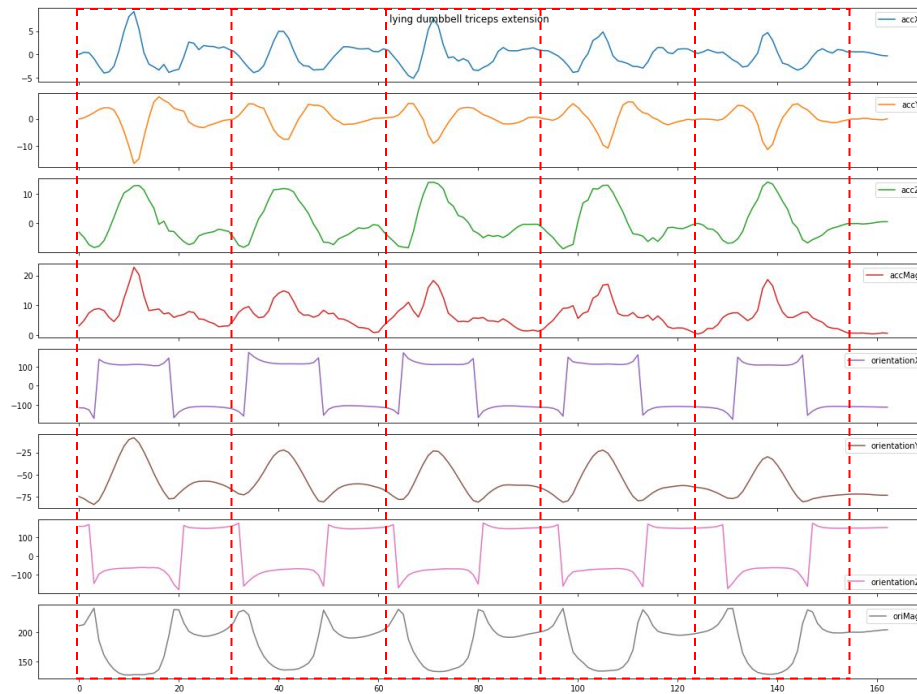
[illegible]

IV. RESULTS / Model Accuracy

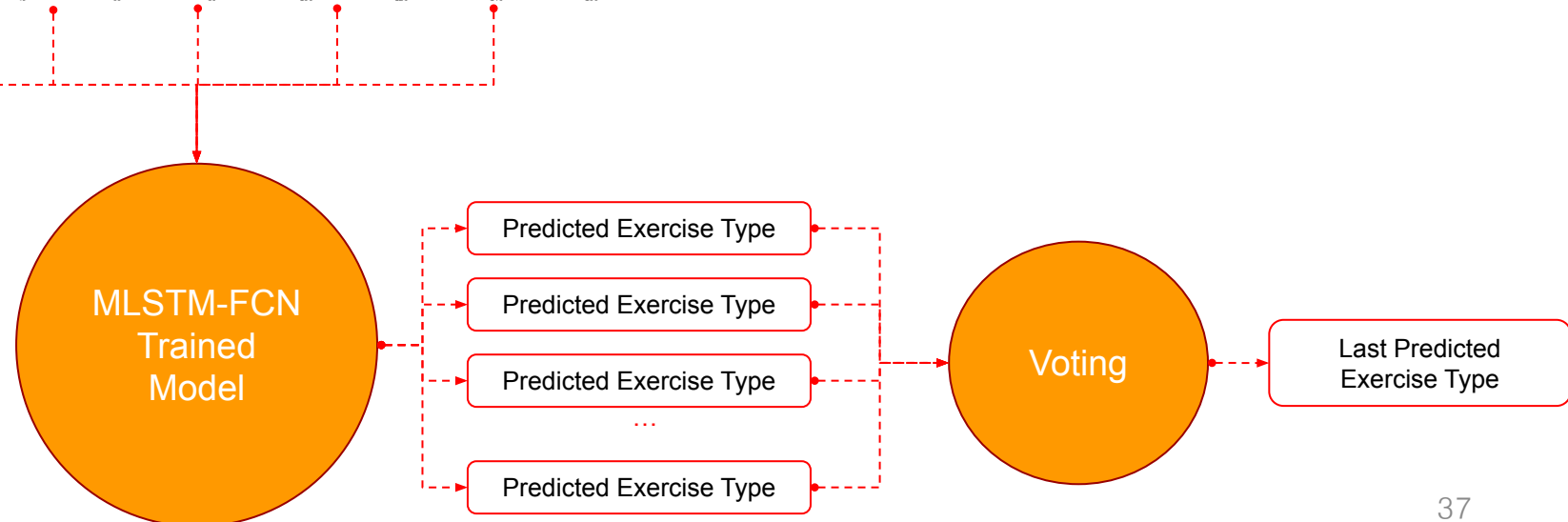
Test Accuracy

[illegible]

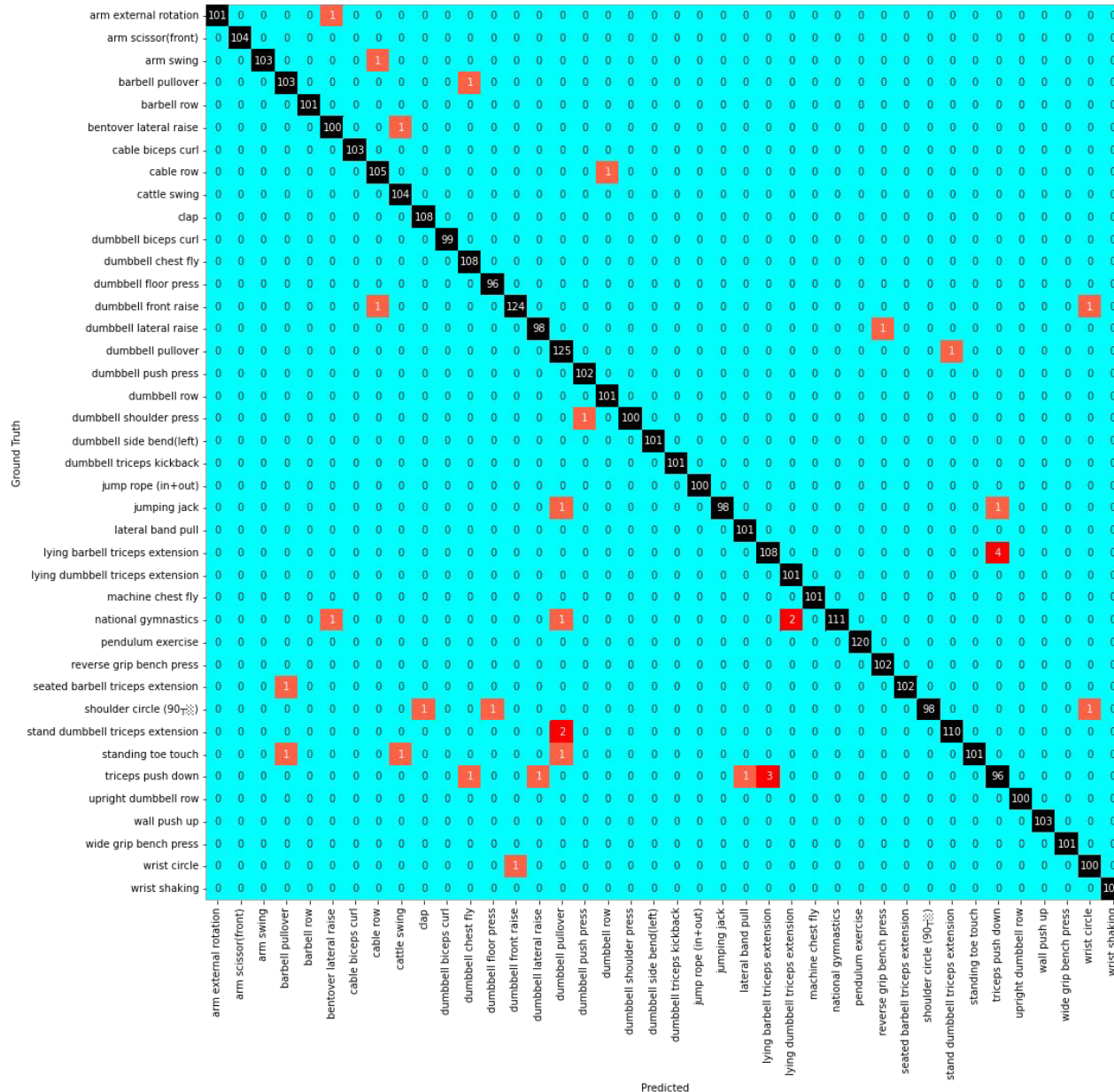
IV. RESULTS / Voting



Each sample corresponds to one of 40 exercise types and they consist of 5 sets. A trained deep learning model tries to recognize the type of exercise based on 1-second (30x8) data from any part of the given sample. In order to increase the accuracy of the model, we split the sample into seconds and predict the exercise type for each second separately by trained model. Finally, we predict the exercise type of the given sample by a majority vote of the model predictions.



IV. RESULTS / Voting



There are total 4182 exercise samples. The sum of each row represents the total number of samples of the exercise type indicated at the beginning of the row. The sum of each column represents the total number of samples prediction by trained model as that exercise type indicated at the beginning of the column.

Main diagonal of the confusion matrix shows number of samples correctly predicted by the model.

The cell i th row and j th column represents the number of samples actually from i -row exercise type predicted as j -column exercise type.

The previous confusion matrix shows that model made only mistake on 10 exercises types. Usually, model should make more than one mistake if the exercise types are similar each other. So, only one error can be caused by the data collection from the sensor. Some samples may have been paused or mistakenly done another exercise. Let's have a look more than one mistakes:

- two **“dumbbell front raise”** samples predicted as **“cable row”** and **“wrist circle”**;
- two **“jumping jack”** samples predicted as **“dumbbell pullover”** and **“triceps push down”**;
- four **“lying barbell triceps extension”** samples predicted as **“triceps push down”**;
- four **“national gymnastics”** samples predicted as **“bentover lateral raise”**, **“dumbbell pullover”**, and **“lying dumbbell triceps extension”**;
- three **“shoulder circle 90° ”** samples predicted as **“clap”**, **“dumbbell floor press”**, and **“wrist circle”**;
- two **“stand dumbbell triceps extension”** samples predicted as **“dumbbell pullover”**;
- three **“standing toe touch”** samples predicted as **“barbell pullover”**, **“cattle swing”**, and **“dumbbell pullover”**;
- six **“triceps push down ”** samples predicted as **“dumbbell chest fly”**, **“dumbbell lateral raise”**, **“lateral band pull”**, and **“lying barbell triceps extension”**;

two “**dumbbell front raise**” samples predicted as “**cable row**” and “**wrist circle**”;

dumbbell front raise



1

cable row



1

wrist circle



two “jumping jack” samples predicted as “**dumbbell pullover**” and “**triceps push down**”

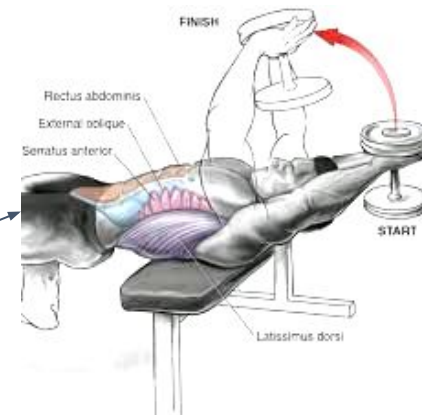
jumping jack

헬스조선



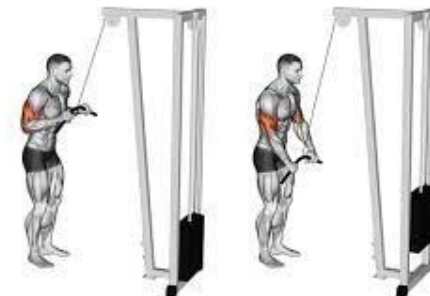
1

dumbbell pullover



1

triceps push down



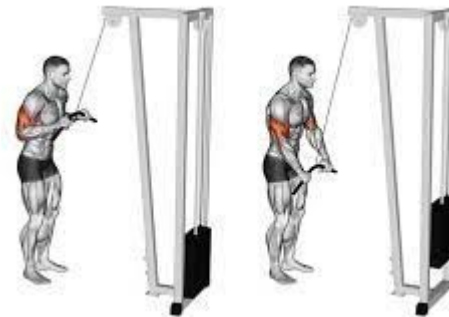
four “**lying barbell triceps extension**” samples predicted as “**triceps push down**”

lying barbell triceps extension



4

triceps push down



four “**national gymnastics**” samples predicted as “**bentover lateral raise**”, “**dumbbell pullover**”, and “**lying dumbbell triceps extension**”

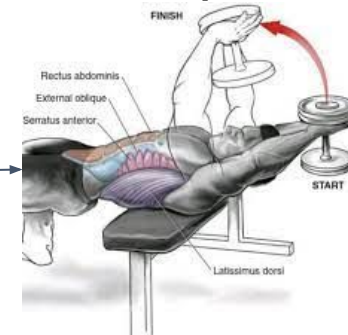
national gymnastics



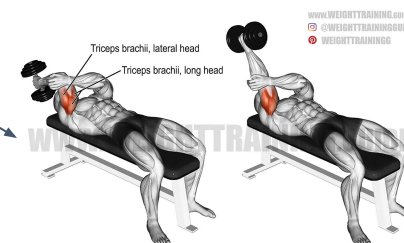
bentover lateral raise



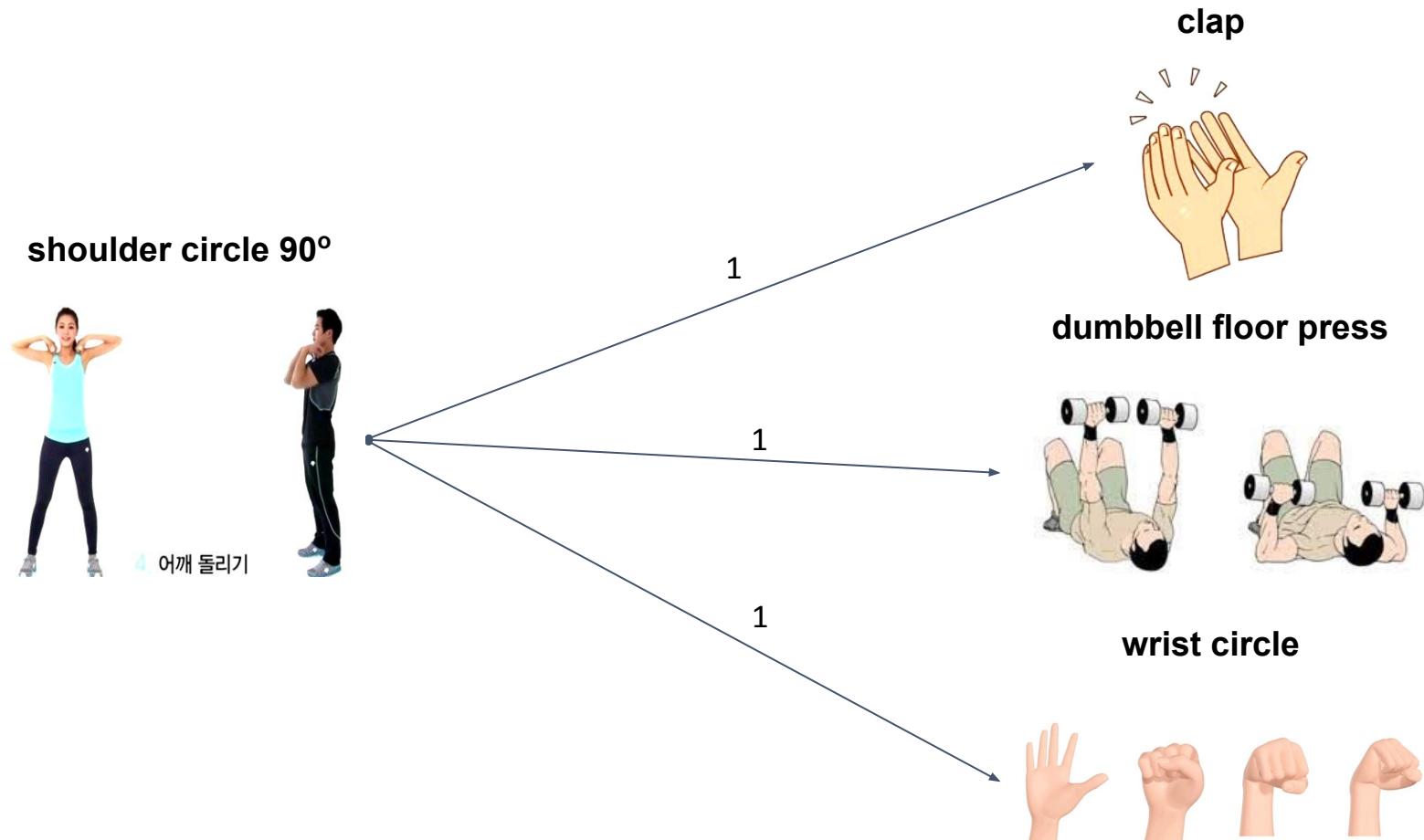
dumbbell pullover



lying dumbbell triceps extension



three “**shoulder circle 90°**” samples predicted as “**clap**”, “**dumbbell floor press**”, and “**wrist circle**”



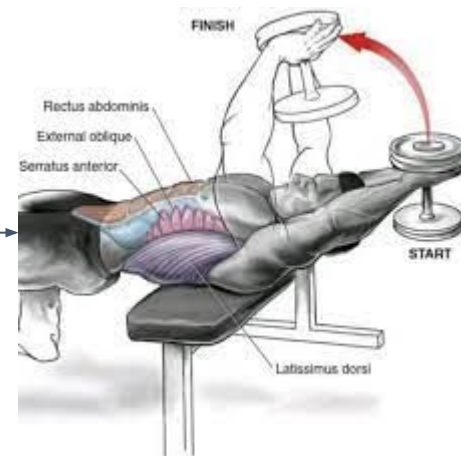
two “**stand dumbbell triceps extension**” samples predicted as “**dumbbell pullover**”

stand dumbbell triceps extension



2

dumbbell pullover



three “**standing toe touch**” samples predicted as “**barbell pullover**”, “**cattle swing**”, and “**dumbbell pullover**”

standing toe touch



1

barbell pullover



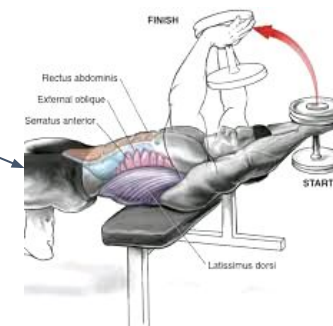
1

cattle swing

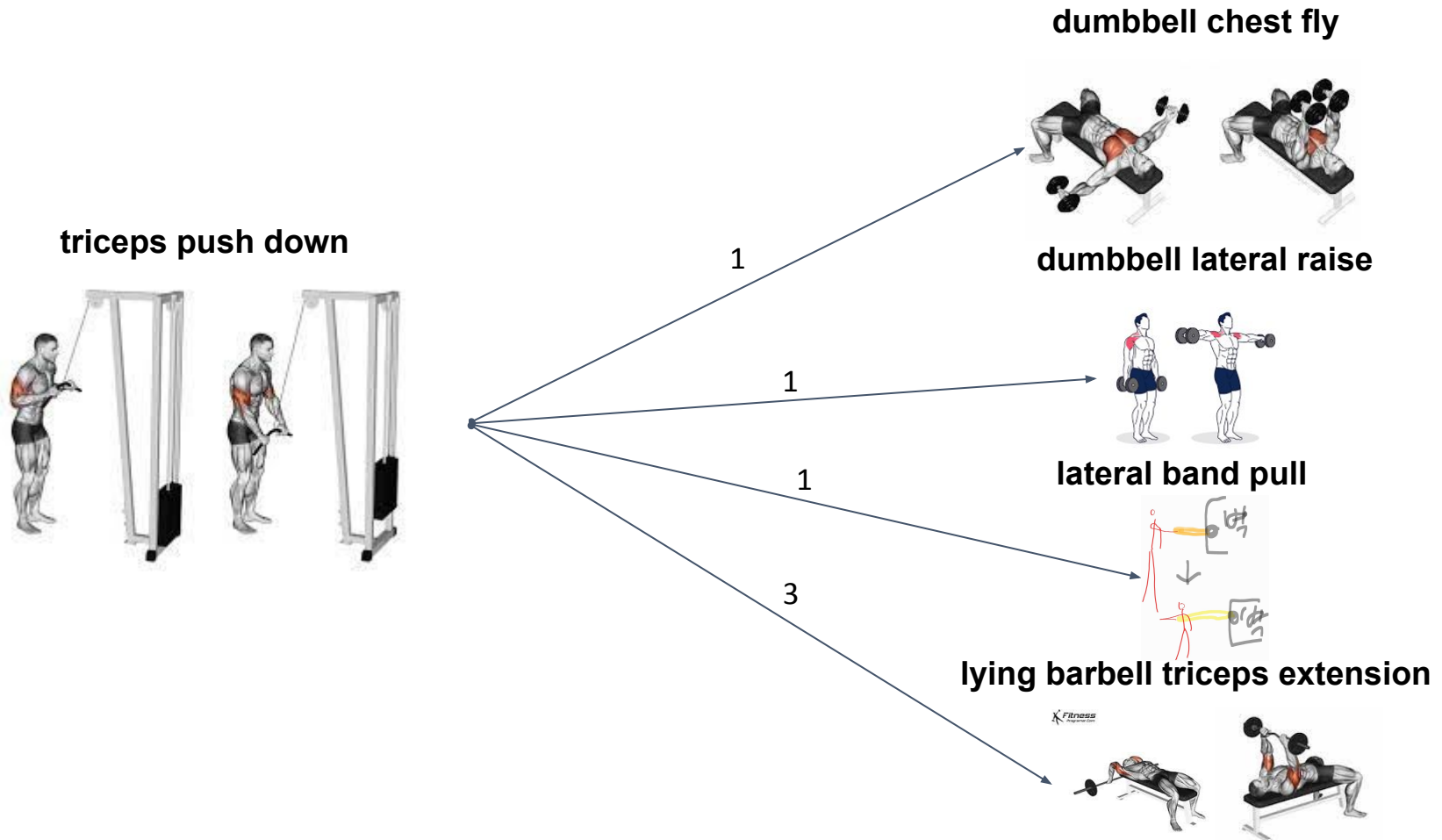


1

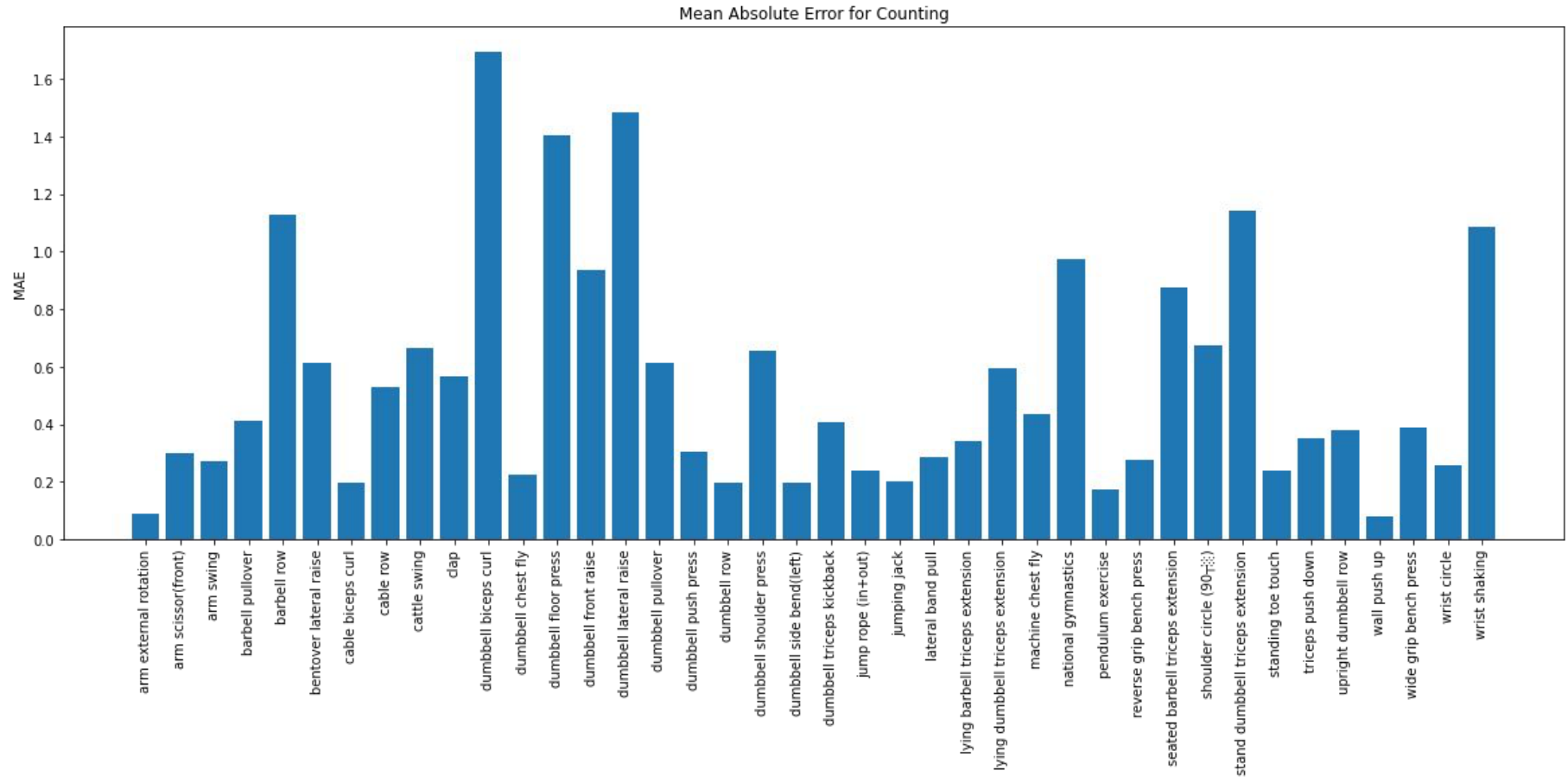
dumbbell pullover



six “triceps push down” samples predicted as “dumbbell chest fly”, “dumbbell lateral raise”, “lateral band pull”, and “lying barbell triceps extension”



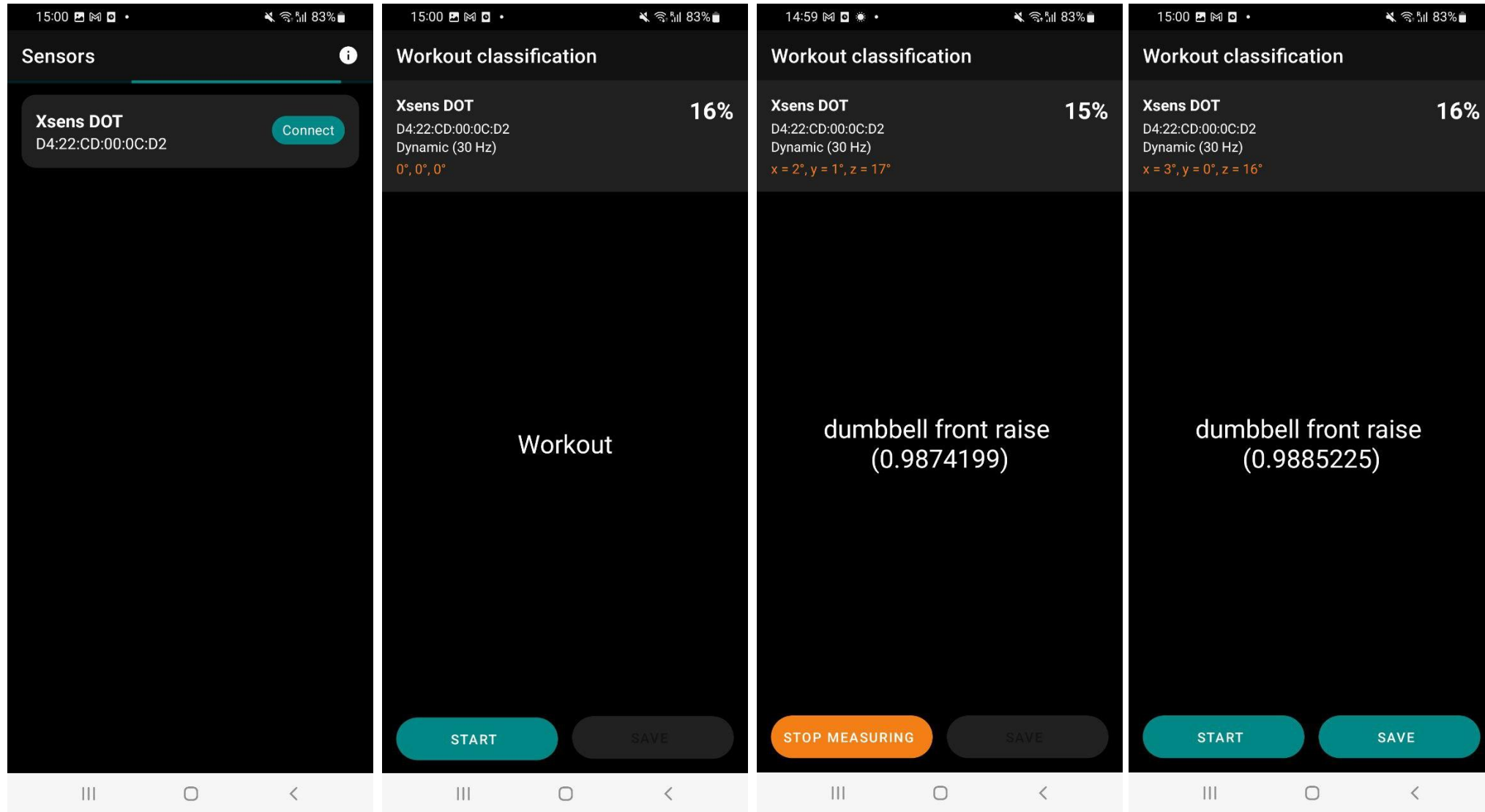
IV. RESULTS / Counting



All 4182 samples were done 5 sets. After predicting the type of exercise, the number of sets of this exercise was calculated by dividing the duration of the exercise by the average time spent performing one exercise of that type. The mean absolute error between actual and predicted number of sets of each exercise samples is 0.55. MAE by each exercise types shown in the bar plot above.

IV. RESULTS / Mobile Application

Real time exercise prediction app - xsens dot sending data to the app real time to predict the exercise type



FUTURE WORK



V. FUTURE WORK / More sensors and more test data for AI training



Simulation





Conclusion. Currently, Our trained Exercise Classification model predicts 40 types of exercises with 98% accuracy even with one “**Xsens DOT**” sensor. There is small mistakes between similar exercises. When counting the set of exercises, the average error is 0.5. It means “**counting**” algorithm may make average ± 1 mistake.

In Future, we going to use 5 set of “Xsens DOT” sensors as shown in the picture left. With 5 sensors,

- We can classify accurately more than 40 exercise types;
- We can simulate exercises in real-time;
- We can improve “counting” algorithm with 5 sensors.
- By tracking each user individually with some other features we can build “**AI Personal Trainer**”.